AN ALMA ARCHIVAL SEARCH FOR SPECTRAL TECHNOSIGNATURES TOWARD STARS WITHIN 40 PC

Glenn J. White[1,2] and Robin Dey[3]
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ABSTRACT

We have searched 431 archival ALMA spectral windows, 89.6–495.1 GHz (Bands 3–8), toward 88 stars in 81 systems within 40 pc for unresolved spectral carriers, comparing each stellar position with 512 controls in an annulus. **No credible technosignature is found.** The sample is every star within 40 pc meeting our public-archive coverage criteria: complete in targets and tunings, not in epochs (§3). It holds 0.50 per cent of that volume's stars, over-representing F and G types and under-representing M dwarfs. Barnard's Star and Wolf 359 gain first millimetre technosignature limits. Four windows are stage-1 spatial outliers. Two toward β Pic and one toward HD 48370 match known circumstellar or foreground CO; the fourth, toward CP−72 2713, has no astrophysical attribution and does not recur in a second, deeper block at the same tuning. It fails the pinned recurrence criterion and is retired, though two epochs cannot separate an excursion from a non-repeating one; the same test recovers β Pic CO in every re-searched block. No flag is significant alone; the test resolves no probability below 1/513, far above the Bonferroni scale. Thresholds are trigger levels, not completeness limits. They span 2.3×10^{13}–6.8×10^{16} W EIRP over the 118 drift-resolved windows and 1.6×10^{13}–2.8×10^{17} W over the 313 coarse ones, against 1 MW on a 12 m aperture at 230 GHz ($\approx 8\times10^{14}$ W). A sub-channel carrier makes them optimistic by $\times2.00$ to $\times2.7$ (median $\times2.29$). Only 118 of 431 windows discriminate drift; the rest search for unresolved spectral excess. The completeness measured for the reference Class A configuration is transferred to the rest by assumption, bracketed 0.5–2×. The search is polarisation-blind, and the drift ceiling does not cover TRAPPIST-1 b, the only sample planet to reach it. The statistic and line mask were defined on these data, so the rank and false-alarm figures describe this dataset rather than independently validating the pipeline's false-positive rate.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

A technological civilisation, signalling deliberately or going about its affairs, may imprint the electromagnetic spectrum detectably, a *technosignature* (Sheikh 2020; Vidal et al. 2026). Six decades of radio searches cover many thousands of stars yet sample only 6×10^{-18} of the haystack of transmitter frequency, bandwidth, power and duty cycle (Wright et al. 2018). Three gaps persist. The millimetre and submillimetre regime is essentially unsearched, and everything above that axis's ∼115 GHz upper limit wholly so. Most surveys are aimed at interest-selected classes of star, leaving the local stellar population unsampled as a population. Few reprocess *existing* interferometric archives.

This paper addresses all three, asking how strongly existing ALMA observations constrain unresolved artificial spectral emission from nearby stellar systems. The sample it uses characterises the stars ALMA has observed, over-representing F and G stars and under-representing M dwarfs (§3), and every limit inherits that bias. We mine the archive, correcting for proper motion and Doppler drift, for unresolved spectral carriers (channel-confined excess at native ALMA resolution) and anomalous millimetre continuum. Throughout, "narrowband" describes the assumed intrinsically narrow transmitter and never the resolving power of the data (§4). Channelisation sets which of two search classes a window belongs to, and the completeness of each class is measured in §4.4. A continuous carrier is the best case and a short-duty fast-drifting emitter the worst, while coarse-channel windows retain sensitivity to persistent unresolved excess power alone. The paper's three contributions are systematic exploitation of archival ALMA stellar fields, 107 target/band datasets mined from public data with no new observing time; spatial-control null calibration for interferometric SETI, the same statistic evaluated at the star and at 512 control positions so the false-alarm rate is measured; and a quantitative accounting of the frequency, EIRP and drift domain actually searched, which gives the non-detection explicit boundaries. No ancillary analysis carries a technosignature disposition. The continuum screen alone feeds an argument in the primary lane (§4.1), and closure-phase vetting, the chirp and periodicity re-analysis and the frequency-occupancy check are capability demonstrations (§4.3). One worked case runs through the paper, a crossing towards β Pictoris, the largest star-versus-control contrast here, attributed to the disc's known circumstellar CO across two transitions and three epochs (§5.3). Per-target results are in Appendix D, and dispositions, false-alarm accounting and injection completeness consolidate in Tables 4 and 5 with Appendices H and B.

The symmetric detection statistic and the $\pm50\,\mathrm{km\,s^{-1}}$ molecular-line mask were both defined, and the statistic once

[1] School of Physical Sciences, The Open University, Walton Hall, Milton Keynes, MK7 6AA, UK; glenn.white@open.ac.uk
[2] RAL Space, STFC Rutherford Appleton Laboratory, Harwell Campus, Didcot, OX11 0QX, UK
[3] VBRL Holdings Inc.

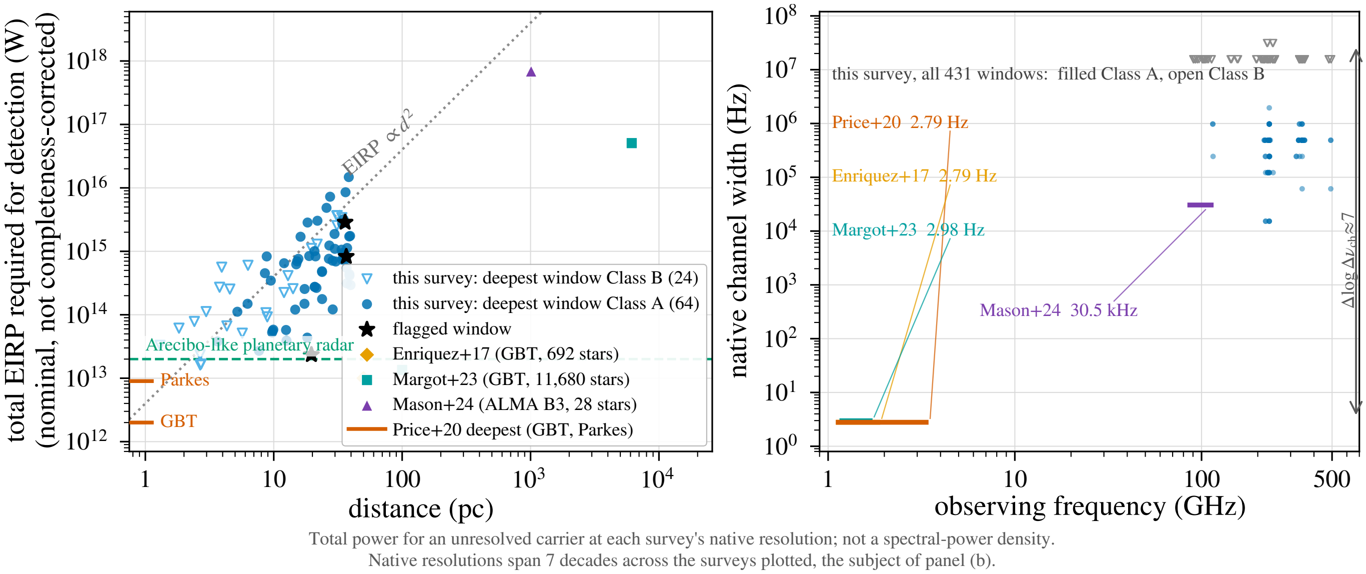

FIG. 1.— Literature context of the 5σ trigger thresholds, in total power (§2). *(a)* EIRP against distance. Blue: this survey, deepest window per star (88 stars, 1.30–39.63 pc; filled: Class A, drift-resolved; open: Class B, unresolved excess); orange: Enriquez et al. (2017); aqua: Margot et al. (2023); violet: Mason et al. (2024); left-edge ticks: deepest per-target limits of Price et al. (2020). Grey dotted: EIRP $\propto d^2$ at constant received flux. Green dashed: Arecibo-class power (§4.2). *(b)* Native channel width against observing frequency for the same windows and programmes: native resolutions differ by $\sim 10^7$, so the panels should be read together and a lower EIRP here does not imply greater sensitivity to a hertz-wide carrier; conventions differ between surveys.

revised, on these data (§5.4). The rank probabilities and false-alarm figures are therefore descriptive of this dataset, not an independent validation of the pipeline's false-positive rate, and this is restated where it bears on a number.

2. BACKGROUND

Radio searches, from Cocconi & Morrison (1959) and Drake (1960) through the programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002), divide into wide-field commensal and targeted surveys (Zhang et al. 2020; Czech et al. 2021; Morrison et al. 2023; Li et al. 2026; Johnson et al. 2023; Valente et al. 2025; Tremblay et al. 2024), with machine-learning interference rejection now central (Ma et al. 2023; Choza et al. 2024; Jacobson-Bell et al. 2025; Poznanski et al. 2025; Garrett et al. 2026), a capability our classical threshold search lacks. Essentially all of it lies below $\sim$10 GHz, and the mm/submm regime above has been searched once, with ALMA (Mason et al. 2024). Physical reasons favour going higher: interstellar scattering and dispersion fall steeply with frequency, a given aperture delivers a tighter beam per watt of EIRP, and directional communication, radar and beamed power are already engineered coherent mm-wave emission. Atmospheric transmission, receiver noise, photons per watt per hertz, and pointing and phase stability raise the cost of such a search without forbidding it.

Mason et al. (2024) searched archival Band 3 windows toward 28 "stellar bycatch" stars, observed for other science and usable for free, placing $\mathrm{EIRP_{min}} > 7\times10^{17}$ W for the nearest and identifying the challenges of high-frequency work: drift rates scale with frequency, smearing erodes sensitivity without a drift search, and the dense forest of molecular lines raises the confusion background. Our pilot (White 2026, citation provisional) extended that methodology to TRAPPIST-1 across Bands 3, 6 and 7 ($\mathrm{EIRP_{min}} \sim 5$–$10 \times 10^{13}$ W, no detection) and built the ranked 100-star list this survey grew from. Nothing here depends on it, because every claim attributed to it below is restated from the products released with this work. The three contributions listed in §1 are new, to our knowledge, *in an ALMA archival technosignature survey*; the four further diagnostics there gate nothing, and we claim no wider priority over their centimetre-wave counterparts.

No execution block is analysed in both papers: the pilot's 100-star search products are superseded here, not re-used, and every window here was re-reduced and re-searched from the raw archive data with the pipeline of §4. Where a star is treated in both, the numbers to cite are this paper's frozen products, and any discrepancy resolves in their favour.

Figure 1 places the nominal EIRP thresholds among published searches in *total power*. As the in-panel note warns, vertical position carries no technosignature merit. This survey's competitive EIRP toward the nearest systems comes predominantly from the d^2 factor (d spans a factor $\sim 10^3$ in d^2), and only weakly from sensitivity to Hz-wide carriers, to which a 15.625-MHz channel and a 2.8-Hz SETI channel respond very differently (§4).

A non-detection must become a quantitative exclusion statement: Wright et al. (2018) formalised the "Cosmic Haystack" search volume, Margot et al. (2023) its conversion into an upper bound on the transmitter-hosting fraction, and Sheikh et al. (2025a) asked where Earth's technosignatures would be detectable. That bound is valid only if the sample represents the underlying population, which interest-selected samples do not. Most surveys select by proximity, spectral type, planets or Earth Transit Zone membership; the ALMA-specific form, disc selection, skews the sample young and dynamically active. We instead take every star within 40 pc lying in the usable field of at least one public ALMA observation (§3), a sample archive-complete for the observed subset of the local population in a fixed volume while making no completeness claim for that population itself. That choice removes technosignature-specific selection and leaves the archival astrophysical selection in place; the 168 sample entries are ∼1 per cent of the catalogued population within 40 pc (the 88 actually searched here are ∼0.5 per cent), and the residual coverage selection is characterised as a completeness function (§3, Table 4).

3. SAMPLE SELECTION

The sample is ALMA-archive-complete within 40 pc, which is a weaker condition than volume completeness. It comprises every star within 40 pc of the Sun (Gaia DR3 parallaxes, distances direct inversions $d = 1/\pi$; every entry $\pi \geq 25$ mas with a parallax measured to better than 20 per cent, which is the condition the archive query applied) whose proper-motion-corrected position at the observation epoch falls in the field of view of at least one public ALMA execution block. The star need not sit at the pointing centre, since mosaics and wide pointings serendipitously cover more than one catalogued star. Disc, planet and spectral type enter as descriptive metadata only. Throughout, a *star* is one catalogue entry and a *system* the set of entries bound to it; a *target/band* dataset is one star in one ALMA band; an *execution block* (EB) one archival observation; and a *window* is one spectral window of one execution block. This gives 168 unique target stars, processed nearest-first. Because trigger thresholds scale with d^2 at fixed noise, that order is sensitivity-ranked, so the prevalence statements of §6 are conditional on it.

Composition follows from the archival selection and is tabulated against the reference census in Table 9. It rests on `teff_gspphot` as a spectral-class proxy, available for 52 of 88 sample stars and missing preferentially for the coolest objects, so the M fraction is a lower limit and the earlier-type ratios upper limits: among searched stars carrying a Gaia temperature, 41 per cent are M dwarfs against 69 per cent of the census. The non-detection therefore bears on an F/G-enriched mix that under-represents the M dwarfs dominating the local volume, a mismatch of astrobiological priority that leaves statistical validity intact. 18 of the 88 processed stars are confirmed exoplanet hosts (§6); the debris-disc hosts (β Pic, AU Mic, τ Cet, and ϵ Eri whose Band 6 windows are withheld, §4.2) contribute dusty continua, kept from masquerading as anomalies by the continuum lane's screen (§4). Window counts are also concentrated: β Pic (15 windows), AU Mic (12) and TRAPPIST-1 (12) alone carry 39 of the 431 searched windows (9.0 per cent) from 3 systems, so any window-weighted statement leans on a few heavily observed young, active or disc-hosting systems.

The 168 stars are complete against, and conditional only on, the 40-parsec reference catalogue intersected with the public-archive snapshot of 2026 September 8. Against that Gaia DR3 solar-neighbourhood catalogue (17 566 stars within 40 pc) they are ∼1 per cent. The two parallax cuts are not interchangeable. The sample's binding condition is $\sigma_\varpi/\varpi < 20$ per cent and the reference census's is ≤10 per cent (Table 9), so the ratio compares two differently selected populations. The realised sample is far better than its condition, σ_ϖ/ϖ reaching 2.48 per cent at worst and exceeding 1 per cent in only 3 entries. The census is also incomplete at both ends, and the fraction shows only that coverage is small. Table 9 leaves the selection function inspectable; age and disc-hosting flags are documented qualitatively, being unevenly available.

Stellar counts throughout are catalogue-entry counts. The 88 searched entries comprise 81 independent systems, differing by seven designation-linked component pairs (SIMBAD designations; per-entry rows, distances and unresolved-pair notes in the machine-readable catalogue). The 107 star/band datasets, 431 windows and 102 execution blocks therefore derive from 81 systems, the paired rows being a bookkeeping convenience. The 431 rows comprise 396 distinct (execution block, spectral window) datasets, and §5.3 and Appendix H use the larger number as the trials denominator, which is conservative.

"Usable public ALMA coverage" is a five-criterion decision tree, frozen absent an erratum: (i) at least one public execution block (EB) whose visibilities have passed ALMA's second-stage quality assurance (QA2) and cover the star's epoch-propagated position in the primary beam, with no minimum integration, sensitivity, antenna count, mosaic-tile count or array configuration beyond QA2; (ii) that position inside the field's primary beam, the extraction aborting beyond $2\,\theta_{\rm PB}$ (largest retained offset $0.46\,\theta_{\rm PB}$, j1256-1257, so nothing approaches the bound; largest retained primary-beam correction 1.81×, §4.2); (iii) each spectral window admitted individually, on QA2 vetting alone; (iv) no programme-category restriction, science, calibration and observatory projects entering on identical terms; (v) a fixed archive snapshot date (2026 September 8). No star or window was excluded by an unlisted criterion, none for short integration (the shortest retained are 1.0 and 1.5 min).

Archive-completeness holds in targets and tunings. It does not extend to epochs or to integration time, and the shortfall is quantified here. The 102 member observing unit sets (MOUS) that supply the searched blocks hold 448 public execution blocks between them, of which 102 were searched (22.8 per cent) and 346 were not. Two deliberate choices cause that, a per-target download budget capping the blocks taken at 3, and spectral-window de-duplication by identifier across the blocks of one set. 75 of the 102 sets hold at least one unsearched block, supplying 304 of the 431 windows (71 per cent) and 62 of the 88 stars, for which up to 34 public blocks exist against the 3 this survey searched. Every epoch statement here, the epoch-activity bound of Appendix J included, therefore concerns searched epochs, and the archive would support a stronger one. 4 of those blocks have since been searched, as the recurrence tests of §5.3. They are targeted follow-ups outside the survey, leaving the counts in this section and the trials budget of §5.3 unchanged.

ALMA project metadata quantify pointed versus serendipitous coverage: almost every spectral-window-covered star is the pointed science target in at least one execution block, leaving Wolf 219 the sole serendipitous star. The residual selection function of an archive-complete sample is the union of ALMA target lists, quantified here at a bycatch fraction of 1 star in 44 and 1 EB in 57.

4. DATA AND METHODOLOGY

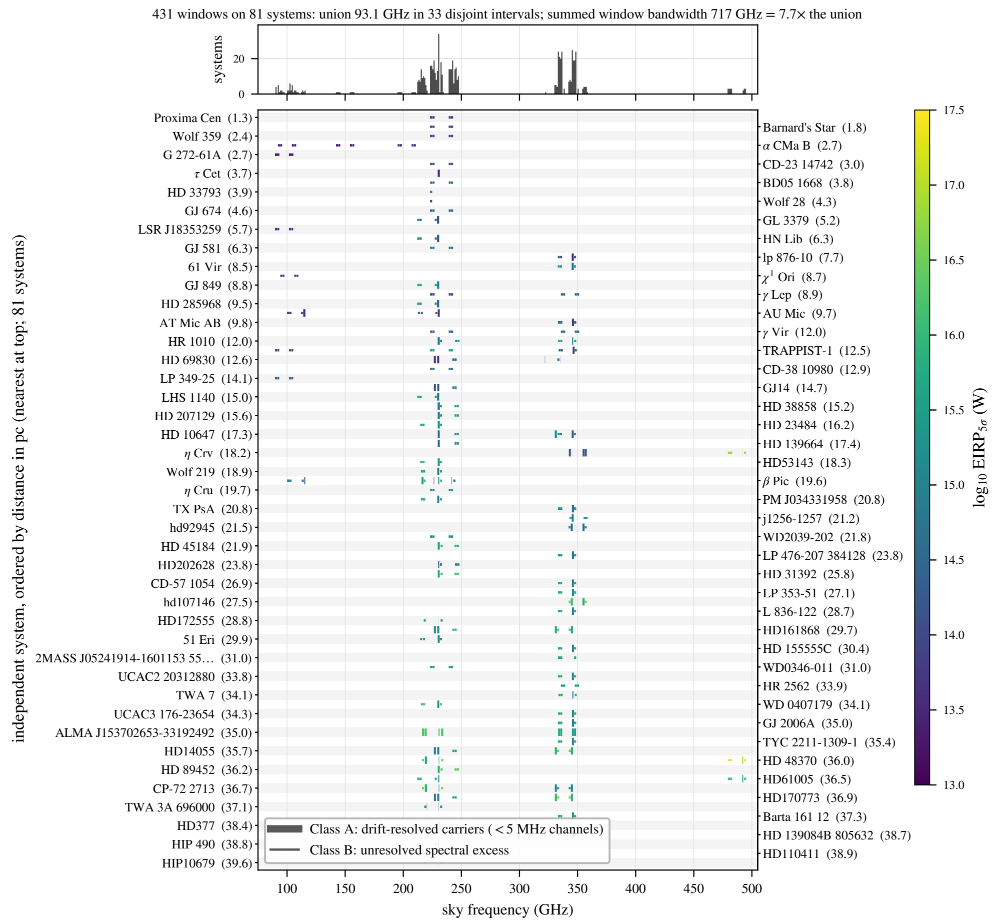

Fig. 2.— What this survey sampled in frequency. Each searched window is a horizontal segment on its host system's row, rows ordered by distance, coloured by nominal EIRP$_{5\sigma}$, with line weight distinguishing the drift-resolved (Class A) from the unresolved-spectral-excess (Class B) channelisation; the upper panel counts systems per gigahertz of sky frequency. The coverage is narrow, clustered and repeatedly revisited: 33 islands concentrated near 230 and 345 GHz, with summed window bandwidth 7.7× the union, which is what the in-band transmitter fractions of Appendix J condition on (Scope of the constraints, §4).

The design is one shared archive-ingestion stage feeding two independent search lanes over the 40 pc archive-complete sample; Table 2 gives the data path of the primary lane in order. For each target we mine, within a bounded budget, all calibrated (or locally recalibrated) archival execution blocks covering the star's Gaia-propagated position. Their visibilities feed (i) the *primary* lane, a Doppler-drift-corrected spectral-carrier search following Mason et al. (2024) and Enriquez et al. (2017), which converts each non-detection into an equivalent isotropic radiated power limit $\mathrm{EIRP}_{5\sigma} = 4\pi d^2 S_{\min}\Delta\nu$. Here $S_{\min}$ is five times the searched window's per-channel rms, primary-beam corrected, $\Delta\nu$ the native channel width and d the target distance (worked numerically in Appendix A). Their systematic budget is the ALMA absolute flux scale (5–10 per cent, band- and epoch-dependent), residual atmospheric phase after water-vapour correction and phase referencing (a coherent point-source loss $\exp(-\sigma_\phi^2/2)$, typically 5–20 per cent at 230–345 GHz, common to the star and the annulus within the isoplanatic patch and so no threat to exchangeability), the primary-beam model at the stellar offset, the instrumental spectral response (the boxed rule below; median ×2.29) and visibility calibration. Each multiplies $S_{\min}$ and hence EIRP$_{5\sigma}$ directly; *Gaia* distance errors are negligible here. The same visibilities feed (ii) an *ancillary* line-excluded continuum lane (Appendix G). "5σ" throughout is the local instrumental *trigger* threshold of Table 1, estimated empirically (§5.3). In most windows an intrinsically Hz-wide carrier is unresolved within a 15.625 MHz channel, and the data could not show that a one-channel excess is intrinsically narrow. The search is reported as two survey classes, distinguished by the dimensionless drift parameter

$$\eta_{\rm drift} \;=\; |\dot\nu_{\max}|\,T\,/\,\Delta\nu_{\rm ch}, \tag{1}$$

the channel-width fraction the *maximum* searched drift $|\dot\nu_{\max}|$ traverses over the full on-source track T at native channel width $\Delta\nu_{\rm ch}$. The track is the right time-scale, the de-drift step accumulating coherently along it; $\eta_{\rm smear}$ (Appendix A) is the same ratio taken over one integration, and the two are not interchangeable. A window is *drift-resolving* when $\eta_{\rm drift} \gtrsim 1$, carrying a carrier across enough channels for its motion to be measured. The drift ceiling inside $\eta_{\rm drift}$ is a search-grid boundary, the widest trial drift the stack evaluates, not a prior on transmitter drift rates; carriers drifting faster are missed, and §6.1 gives the resulting selection function (Fig. 7).

Class A is the drift-resolved spectral-carrier search: the 118 windows of fine channelisation, native widths 15.259–1953.125 kHz (median 488.3 kHz; 62 per cent at the 488.3-kHz mode), spanning $\eta_{\rm drift} = 0.34$–422 (median 12.8). It is

TABLE 1
Nomenclature for search output and the two senses in which a probability is quoted.

Term	Meaning
crossing	one channel×drift cell at the stellar position with $T \geq 5$: a threshold crossing (state 1)
stage-1 spatial outlier	a window whose $T_\star$ exceeds all 512 control statistics ("star-exceeds-ring"), a screening device that carries no candidate significance (state 2)
$p_{\rm rank,min}$	$1/(N_{\rm ctrl}+1) = 1/513$, the rank *resolution* of one window's control ensemble: the smallest per-window probability the design can express
P(false flag)	the survey-level false-positive probability after thresholding, correlation and non-Gaussianity. The two differ, the first being a property of one window's ensemble and the second of the whole survey, and the survey-wide Bonferroni scale 1.2×10^{-4} lies two orders of magnitude below the first (§5.3)

the only class with measured end-to-end recovery of a drifting carrier. *Class B* is the unresolved spectral-excess search: the 313 coarse windows, 73 per cent of the sample, native widths 15.625–31.25 MHz (309 windows at the 15.625-MHz mode), spanning $\eta_{\rm drift} = 0.004$–1.35 (median 0.36). It is sensitive to persistent near-stationary excess power, carries no drift discrimination, and is never described as a narrowband search anywhere in this paper.

$\eta_{\rm drift}$ is the physical criterion and the classes are labelled by the instrument. Channelisation is bimodal, with no window between 1953.125 kHz and 15.625 MHz, an empty gap of factor 8.0, so any boundary inside the gap assigns identical membership and figures label the classes by name. The two criteria do not coincide exactly, and the overlap is quantified: 12 long-integration Class B windows reach $\eta_{\rm drift} \geq 1$, and only marginally (maximum 1.35), where a one-to-two channel traverse supports no discriminative drift fit after baseline removal; 6 short Class A windows fall below unity and are conservatively kept in Class A. Each window's $\eta_{\rm drift}$, $\eta_{\rm smear}$, trial drift count and maximum searched acceleration $a_{\rm max} = 3.6$–$4.0\,{\rm m\,s^{-2}}$ are columns of the released catalogue, so any reader may reclassify on the physical criterion alone. Results are given per class first and combined only where statistically appropriate (§5.3).

Both lanes exclude catalogued molecular transitions at tolerances suited to their statistics. The continuum lane masks the common species at native resolution before integrating; the carrier lane applies a $\pm 50\,{\rm km\,s^{-1}}$ stellar-frame velocity exclusion mask (§5.3) to any threshold crossing. The mask is *excluded search space*: the unmasked survey covers 93.1 GHz of unique sky frequency, the effective default quoted throughout is the ≈88.2 GHz surviving it, and no constraint is placed on carriers inside the excluded velocity regions. Masking reduces the frequency space a candidate can occupy while leaving the per-channel noise untouched, so no threshold moves. All flux densities and EIRP thresholds are total intensity (Stokes I), summed over the two parallel-hand cross-correlations with QA2 weights; no cross-hand extraction or circularity test was performed, so the construction is polarisation-blind in amplitude and a strongly polarised artificial signal responds like an unpolarised one at fixed total power. The estimator is safe in this respect, though the data have not been audited for it. The retained products do not record per-correlation flagging, so we cannot say what fraction of datasets kept both hands unflagged through QA2; one that lost a hand would be $\sqrt{2}$ less sensitive than its threshold implies, with nothing here to catch it.

How to read every limit in this paper. *Every EIRP$_{5\sigma}$ value here is a nominal, local instrumental detection threshold, the* trigger power $P_{\rm trig}$*: five times the per-channel rms of the searched window, at that window's native channel width, for a transmitter at the star's position matching the searched spectral morphology, with a measured recovery rising with the fraction of the track the carrier occupies its channel, highest for a continuous one (§4.3, Appendix B). A calibrated completeness limit is a different quantity, the power P_x at which recovery reaches x per cent. $P_{50} = 1.10\,P_{\rm trig}$ and $P_{90} > 2\,P_{\rm trig}$ for the calibration configuration's drifting class, both carrying an* unbounded *transfer systematic to any other configuration (bracketed as* 0.5–2×*, Appendix J), where they are* not determined*; $P_{\rm trig}$ is quoted throughout. ALMA's native channelisation sets it, so it is a total-power threshold and no measure of Hz-resolution sensitivity, optimistic by* ×2.00 *to* ×2.7 *under the instrument's Hanning response (Appendix A). Every headline EIRP is quoted beside its native channel width, and the machine-readable catalogue carries the nominal threshold and both Hanning-corrected values per window. The same reading applies to every subsequent "5σ".*

Three signal morphologies are detectable here, and §6.1 quantifies each with real windows. A continuous frequency-stationary carrier is the best case, recovered with near-ideal gain toward the nearest systems. A continuously drifting carrier in fine channels is recovered in 42 per cent of injections at threshold and 75–83 per cent at twice threshold. An intermittent carrier of duty cycle 1/4 is recovered in 91 per cent of fine and 100 per cent of coarse windows.

4.1. *The search statistic, and the words used for its output*

The primary statistic is formed on extracted spectra, never on images or visibilities. It operates on the QA2-calibrated measurement set as delivered, per integration and per native channel, carrying the archive's flagging through untouched and applying no spectral regridding or interpolation (Table 2, step 3), and returns the weighted vector average of the phase-shifted visibilities at the propagated sky position $\mathbf{x}$: the natural-weighted dirty-map value there, with no deconvolution, so a quoted $S_{\rm peak}$ in a field with extended emission is a dirty-map amplitude. Baseline removal follows extraction (step 4), and the primary-beam correction enters the quoted thresholds and fluxes (§4.2), leaving untouched the extracted amplitudes the statistic ranks. For a sky position $\mathbf{x}$, each per-integration spectrum $I(\mathbf{x}, t, \nu)$ is baseline-subtracted by a block median along the *frequency* axis. No temporal median is applied at any stage (§4.3). The residual dynamic spectrum is stacked coherently along each trial linear drift rate $\dot{\nu}$ with inverse-variance weights, and the statistic is the maximum of that stack over the channel×drift grid,

$$T(\mathbf{x}) = \max_{\dot{\nu},\, \nu} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot{\nu} t)\,/\sigma^2(t,\nu)}{\left[\sum_t 1/\sigma^2(t,\nu)\right]^{1/2}}, \tag{2}$$

a matched filter whose denominator is the noise of the stacked spectrum. Only when σ is the same for every integration and channel does it reduce to a single-channel amplitude divided by one noise scale; otherwise the weights vary along the track. The scale $\sigma(t, \nu)$ is 1.4826× the median absolute deviation taken *across the control positions* at each integration t and each channel ν, which makes it global in position, local in frequency and in time, and shared by the star and by every control.

The across-position median enters only in forming that deviation and is never subtracted from the numerator, so T is an absolute stacked signal-to-noise carrying a cross-position noise scale, and it measures no local contrast. A line filling the primary beam therefore drives star and annulus up together instead of cancelling.

The baseline filter takes the median in contiguous blocks of 65 channels and interpolates linearly between block centres, with $\max(2, n/65)$ blocks over n usable channels, so its effective width is $W = 65$ channels for the 118 windows holding more than 130 channels (65–74 in practice) and $n/2$, or 30–59 channels, for the short coarse windows whose block count collapses to two. The linewidth ceiling that imposes is ≈32 MHz at the fine class's modal 488 kHz channelisation and ≈0.9 GHz at the coarse class's 15.62 MHz (Appendix A). The crossing channel is *not* excluded from the scale. The median is taken across positions at a fixed channel, so what matters is how many of the 512 controls carry the feature. A compact feature at the star appears in none of them and the 50-per-cent breakdown point holds; emission filling the primary beam appears in all of them, which is the HD 48370 case (§5.3). $T_\star \equiv T(\mathbf{x}_\star)$ at the proper-motion-propagated stellar position, and the same operation, drift grid, channelisation and noise map give $\{T_i\}$ at all 512 control positions.

The controls are placed by `probe_positions()` in the released extraction code, which adds its offsets to the star's own direction cosines, so the annulus is centred on the *star* itself, wherever the star sits relative to the field phase centre. It holds 512 positions drawn uniform in area, $r = \sqrt{U(r_{\rm in}^2, r_{\rm out}^2)}$, and uniform in azimuth, from the fixed seed 20260825, so the layout is reproducible. Radii run 0.14–0.78 $\theta_{\rm PB}$ with $\theta_{\rm PB} = 1.22\,\lambda/D$ evaluated at each window's own median frequency, so the annulus scales window by window. A separate inner check-ring of 135 positions, reaching 0.06 $\theta_{\rm PB}$, serves the point-source test. The full per-window geometry ships as catalogue columns (Data Availability).

Because the frozen products carry none of it, the geometry is reconstructed from the code, and two checks confirm the layout that ran: the inner check-ring it generates has the 135 positions the retired region-max statistic maximised over, and where extended CO fills the beam the control statistics fall with reconstructed radius (Spearman $\rho = -0.49$, $p \sim 10^{-32}$, in the HD 48370 Band 6 window). A wrong position-to-index mapping could not produce either.

Placing controls off-centre costs sensitivity. A star near the pointing centre sits at primary-beam response ≈1, while the annulus samples 0.95 at 0.14 $\theta_{\rm PB}$ falling to 0.19 at 0.78 $\theta_{\rm PB}$, area-weighted mean 0.47 (lower on ALMA's actual 1.13 λ/D beam). Star and controls are therefore not exchangeable in flux density, while they remain exchangeable in the signal-to-noise ratio the statistic actually ranks, because $T(\mathbf{x})$ in Eq. 2 is formed on uncorrected amplitudes against one noise scale shared by every position. Extended sky brightness does not cancel, and the interferometric mechanism is specific. Smooth emission lives on short baselines, whose phases barely change under a phase rotation of a few arcsec, so its extracted amplitude is nearly position-independent inside the beam and star and annulus rise *together*, as they do for HD 48370. Emission whose centroid is offset from the star instead raises particular controls above it, as in the same star's ^{13}CO window, whose ring maximum of 20.4 at 9.2 arcsec stands against 4.6 at the star (§5.3). Two asymmetries remain, in opposite directions. A suppressed control can only make a star-exceeds-ring outcome more likely under the null, while a multiplicative calibration error scales with the continuum flux at the extracted position and so favours the star, where that flux is not zero. The second is bounded below, and the bound is given with the noise estimator above.

Neighbouring controls are correlated below the synthesised-beam scale, so a window's annulus carries $N_{\rm eff} \simeq 30$–43 independent spatial trials in the compact configurations where the symmetric reprocessing runs measure it, which is the worst case. On each window's own archived synthesised beam the annulus *contains* $N_{\rm geom} = \pi(r_{\rm out}^2 - r_{\rm in}^2)/1.133\,\theta_{\rm beam}^2$ resolution elements, median 995 and reaching 1.4×10^5 (Appendix H). That counts what the annulus could support, never what was drawn, so outside the most compact configurations the 512 controls are effectively independent and $N_{\rm eff} \to 512$. The rank test needs exchangeability rather than independence in any case, and correlation among controls costs resolution without touching the 1/513 floor. The frozen products do not carry the synthesised beam or the array configuration, so radii are quoted in arcsec and in $\theta_{\rm PB}$; the ALMA archive carries both, and what they say about this geometry, including the 33 per cent of windows taken with the ACA 7 m array against a $\theta_{\rm PB}$ defined on 12 m, is in Appendix H.

Ranking $T_\star$ against 512 values fixes the smallest per-window p-value the design can express at 1/513, while a survey-wide Bonferroni scale over 431 windows is 1.2×10^{-4}, two orders of magnitude smaller. **No single event**

TABLE 2
The statistical data path, in order. Steps 1–8 run identically at the stellar position and all 512 controls; only step 9 onward distinguishes them.

1	calibrated measurement set from the ALMA archive
2	stellar position propagated to the observation epoch (Gaia DR3)
3	spectral extraction at that position, native channelisation
4	per-integration baseline removal: block median along frequency ($W = 65$)
5	stack along each trial linear drift rate
6	robust noise map $\sigma(t, \nu)$: MAD across the 512 controls
7	$T(\mathbf{x})$ from Eq. 2, an inverse-variance-weighted stack
8	repeat 3–7 at 512 control positions on a ring
9	threshold: $T_\star \geq 5$ (a crossing)
10	rank test: $T_\star > \max_i T_i$ (stage-1 spatial outlier)
11	molecular-line mask, $\pm 50\,\mathrm{km\,s^{-1}}$ stellar frame
12	vetting: recurrence, closure phase, cross-target occupancy

in this design can therefore reach survey-wide significance. A stage-1 flag selects a feature for examination and establishes no statistical significance. The annulus estimates the local false-alarm rate; it cannot authenticate an individual event (component v, §5.3). The vocabulary for search output nests strictly (Table 1), and Table 2 allows the statistic to be reimplemented from the main text alone.

Because σ is built from the annulus, it carries no variance that belongs to the stellar position alone: residual continuum after baseline removal, deconvolution or self-calibration residuals, and position-correlated calibration error all concentrate at or near the phase centre, where the pointed science target sits. The continuum lane detects 17 of 57 target/bands (Appendix G), so these stars are not all blank there. A multiplicative spectral calibration error carries the same asymmetry and in the unfavourable direction: a residual bandpass ripple of fractional amplitude ϵ leaves a channel-confined artefact of order $\epsilon\, S_{\rm cont}$ at the extracted position and essentially nothing on the annulus, where the continuum flux is negligible. The requirement is severe where it matters, and §5.3 works it through for the one window where it was tested. The release does not carry per-target continuum flux densities, so that argument scales from typical fluxes and bounds no individual window. A direct comparison of the star's own residual variance against the controls' would be the sharper test. The release retains full dynamic spectra for the star and for only 8 of the 512 controls per flagged window, too few to calibrate a variance ratio, and this is the one gap these products cannot close. What they do allow is a measurement of the scale's frequency dependence. Re-extracting the four flagged windows' per-integration spectra and recomputing the scale within a band around the feature and excluding the feature itself, gives local-to-global ratios of 0.97 (CP−72 2713), 1.29 (HD 48370), 0.93 (β Pic B3) and 1.02 (β Pic B6). The global scale is accurate to a few per cent except towards HD 48370, where the local value sits 29 per cent higher because that field is filled with CO; rescaling both the statistic and its ensemble by the local value leaves it flagged ($T_\star = 21.0$ against a corrected ring maximum of 20.8), the disposition already assigned to it. For CP−72 2713 the local scale is slightly *lower*, so its excess is if anything understated and no disposition moves. The retained products keep no drift-stacked cube, so this tests only the noise scale, and a per-crossing local scale inside the search is an open item (§6.3). The same estimator runs at the star and at all 512 control positions, so a badly misstated σ would displace the control distribution, and the pseudo-star ranks are uniform in every band and at both channelisations (§5.3). The one departing stratum is the fine-channel one, and that departure is the four flagged windows themselves.

A rank means something only when the identical operation runs at the star and at every control. Maximising over an $n_{\rm src}$-position region at the star while comparing against single-position controls breaks that requirement. Under a pure-noise null the region maximum exceeds all $n_{\rm ctrl}$ single controls with probability $n_{\rm src}/(n_{\rm src} + n_{\rm ctrl}) = 20.9$ per cent for the 135-position check-ring, an inflation that is a property of the estimator and holds for any window and any noise distribution. The released statistic therefore evaluates one position at the star and one at each control. An earlier version used the asymmetric region maximum; §5.4 states the chronology and audits the windows it affected.

The searched drift-rate range is the larger of (a) a generic frequency-scaled default (12–$13\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$), the Mason et al. (2024) fiducial with a threefold safety margin, and (b) for confirmed exoplanet hosts, the maximum line-of-sight orbital acceleration of the innermost known planet, so a transmitter on a short-period planet cannot drift out of the window. Drift rate maps onto line-of-sight acceleration through the first-order Doppler relation

$$a_{\rm los} \;=\; c\,\frac{\dot\nu}{\nu}, \tag{3}$$

so the ceiling is quoted in cross-band-comparable form $\dot\nu/\nu$: 12–$13\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$ is $\dot\nu/\nu = (1.2$–$1.3) \times 10^{-8}\,\mathrm{s^{-1}}$, i.e. $|a_{\rm los}| = 3.6$–$3.9\,\mathrm{m\,s^{-2}}$ (Eq. 3), independent of observing frequency. That is the acceleration at $\sim$0.04 au around a solar-type star. Earth-motion contributions ($\lesssim 0.13\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$) lie two orders of magnitude below, subsumed by the grid. The grid itself is uniform in $\dot\nu$, spanning the searched range in $n_{\rm drift}$ steps (2–1944 across windows, recorded per window in the released file), so the worst-case mismatch, a carrier midway between trial rates, leaves a residual drift that traverses at most 0.26 of one native channel over the longest fine-class track (0.29 over the coarse class), and the stacked-amplitude loss is bounded by that sub-channel smearing alone, an order below the recovery slope measured between 5σ and 10σ. The injection campaign injected on grid points; continuous randomisation between them belongs with the stratified campaign of §6.3. Very short-period systems lie at the edge of the searched drift range. An artificial emitter need not be tied to a known planet at all, since a free-flying platform, rotating structure or deliberately chirped carrier can accelerate as it chooses, and such emitters fall outside the constraints reported here without being excluded by them. These are *apparent* topocentric drift rates, including deliberate transmitted chirp and acceleration-induced drift, which the survey does not distinguish, so the constraints concern linear apparent drift within the searched range only. For circular orbits the ceiling covers a transmitter co-orbiting an Earth analogue by three orders of magnitude and Proxima b comfortably, while TRAPPIST-1 b ($a_{\rm los} = 4.00\,{\rm m\,s^{-2}}$) sits at it and c ($2.14\,{\rm m\,s^{-2}}$) reaches it only at modest eccentricity. Eccentricity raises the periastron value by up to $(1+e)/(1-e)^2$, $1.9\times$ at $e = 0.2$, so borderline cases move outside the searched domain at modest eccentricity (Fig. 7).

Across the 431 windows maximum searched drift rates span 1.1–5.9 $\mathrm{kHz\,s^{-1}}$ and trial drift rates per window 2–1944 (median 4). The threefold margin matters. At $1\times$ the search would have been blind to two of its threshold-crossing windows, including the AU Mic crossing, while at $5\times$ no disposition changes; transmitters drifting faster than the 12.0–13.3 $\mathrm{Hz\,s^{-1}\,GHz^{-1}}$ ceiling remain unconstrained. Intra-integration smearing is negligible for 422 windows: at most 1.10 channels, median 0.001. 9 windows have $\eta_{\rm smear} < 0.99$ (Appendix A); 5 of them, all 15.3 kHz Band 6 windows and so among the most drift-capable here, carry effective thresholds 1.65–1.74$\times$ nominal, and 2 are stage-1 flags. Channel widths are effective resolutions from the measurement sets' spectral window tables (Appendix A).

4.2. *Frequency and velocity reference frames*

Every frequency axis here, searched spectra, tabulated ranges and crossing frequencies alike, is *topocentric* as delivered by the ALMA correlator, with no velocity-frame shift applied at any stage. Velocities enter only in the vetting stage's line mask, whose conversion chain is in Appendix A. The stacking is coherent in frequency and in the drift-rate trial, and preserves no electric-field phase coherence between integrations.

The native channel width is also the linewidth convention for every EIRP threshold, the commonest confusion when technosignature limits are compared across surveys. Channel dilution spreads a sub-channel carrier's total line flux across the channel, so the minimum detectable total power is $4\pi d^2 S_{\rm min}\Delta\nu_{\rm ch}$, independent of the transmitter's own width below the channel (Appendix A). $\mathrm{EIRP}_{5\sigma}$ is thus the minimum *total* power of a channel-confined carrier, never a spectral power density. Dividing by $\Delta\nu_{\rm ch}$ gives the minimum spectral power for wider transmitters, so a 10^{15} W threshold at 15.625-MHz channelisation is $6.4\times10^{7}\,\mathrm{W\,Hz^{-1}}$. The 1-Hz and 3-Hz limits of Hz-resolution surveys are total powers in the same sense, each the best case of its own channelisation, which makes the axes of Fig. 1 commensurable. Channelisation is the dominant price the archive exacts, and we do not project an Hz-resolution sensitivity from it, because a radiometer-equation scaling $P_{\rm min} \propto \sqrt{\Delta\nu}$ across six orders of magnitude in spectral resolution would be a thought experiment not an achievable ALMA sensitivity.

Every spectral window of an observation enters the spectral-carrier search, capped at 8 per target to bound compute cost. The cap bound one target/band, 61 Vir Band 7 (four of eight windows searched), whose frequency-space selection function is tabulated; three further targets (τ Cet, Kapteyn's Star, Van Maanen's Star) are represented by their deepest single window and are marked in the released catalogue. Each searched window yields its own $\mathrm{EIRP}_{5\sigma}$ row, and Figure 1 plots only the deepest per star. When a crossing is checked against the masked species the operative discriminant is velocity coincidence: circumstellar emission sits at rest offset by the gas's kinematics, while a technosignature may appear at any frequency and drift rate. The searches as executed used a narrower one-channel-width implementation, which agrees with the velocity-space form on every disposition here (§5.3).

Both carrier-lane thresholds and continuum fluxes are corrected for ALMA's primary-beam response at the star's offset from the phase centre, negligibly for most targets (median factor 1.00004, 90th percentile 1.018), but 41 of the 431 retained windows exceed 2 per cent and 31 exceed 10 per cent, across 6 catalogue entries (five designations: the two HD 139084B rows share one). The largest retained correction is j1256-1257 Band 7 at 1.81$\times$, at $0.46\,\theta_{\rm PB}$, and Sirius B spans 3.6–18.6 per cent over its 12 windows at 0.11–0.25 $\theta_{\rm PB}$. The boundary between a retained bounded correction and a withheld window is whether the Gaussian beam form is defensible at the offset in question, and the offset by itself does not decide it. ϵ Eri Band 6 lies beyond the first sidelobe transition, where the correction reaches 2.9–3.4$\times$ and the Gaussian form fails, so we *withhold* those four windows, which enter no number here but are released, flagged, in the catalogue. Everything retained sits inside $0.46\,\theta_{\rm PB}$, where the correction stays a bounded systematic within a valid beam model. Excluding Sirius B would change neither endpoint of the quoted EIRP range nor any conclusion, and would move the median by one significant figure, $1.4 \rightarrow 1.5 \times 10^{15}$ W. The correction uses the same $\theta_{\rm PB} = 1.22\,\lambda/D$ as the annulus geometry, the Airy first-null coefficient rather than ALMA's FWHM of $\approx$1.13 λ/D, so off-axis thresholds are optimistic by up to $\sim$10 per cent at the largest retained offset, which is 0.50 of a true FWHM.

The EIRP thresholds are compared throughout to an Arecibo-like planetary radar (Drake 1974; Sheikh et al. 2025a, EIRP 2×10^{13} W), which is a technological *power* scale carrying no transmitter model. It is an 2.38 GHz number, and aperture gain scales as ν^2, so the same aperture and transmitter power at 230 GHz would radiate $2\times10^{13}\times(230/2.38)^2 \simeq 2 \times 10^{17}$ W. The Arecibo figure carries that dish's real aperture efficiency; the frequency-matched benchmark below is geometric. That benchmark is a 12-m aperture radiating 1 MW at 230 GHz, EIRP $\approx 8 \times 10^{14}$ W, within a factor of two of the median Class A threshold 7.5×10^{14} W. Neither benchmark implies a transmitter model.

4.3. *Ancillary and validation analyses*

Four further analyses are summarised here as far as the main text depends on them.

Injection-recovery validation (Appendix B). Synthetic tones of known amplitude were injected into real visibilities and recovered with the unmodified pipeline: 1200 trials across six window configurations, three bands and both resolution classes. The pipeline's baseline step is a per-integration block median along *frequency* (Table 2, step 4), which by construction cannot remove a feature confined to one or two channels, and its de-drift step is an inverse-variance-weighted sum over integrations, which accumulates a persistent carrier coherently. Appendix B tells one validation-history item in full. The campaign's original report of 0 of 500 coarse-window recoveries was an artefact of its recovery *criterion* and never of the search pipeline. The corrected criterion has not been re-derived by an independent implementation, and that cross-check is an open item (§6.3). The independent evidence is the dwell campaign of §4.4, which measures persistent near-static carriers as *recovered* (Table 3), reversing the artefact's reading.

The primary statistic has no automated injection test on every pipeline change; three quantified checks bound undetected pipeline faults instead: the machinery-only lane (fidelity $-9/+14$ per cent, the bound that caught this artefact), the AU Mic re-analyses of §5.4 under three treatments, whose recomputation agrees with the released products, and the pseudo-star rank uniformity (220 672 pooled ranks uniform, mean 0.5010 against 0.5, KS $p = 0.37$), which any defect of the statistic itself would displace. For the drifting class on the fine window, recovery is 17, 42, 58, 75 and 83 per cent at 4, 5, 6, 8 and 10σ, measured on the single calibration configuration of one star and one Band 6 window and transferred by assumption to all 118 Class A windows, an error the bracketing of Appendix J treats as a 0.5–2× rescaling. That gives $\mathrm{EIRP}_{50} = 1.10\,\mathrm{EIRP}_{5\sigma}$ and, as a bound, $\mathrm{EIRP}_{90} > 2.0\,\mathrm{EIRP}_{5\sigma}$, each with an unbounded transfer systematic bracketed as 0.5–2× (Fig. 6, Table 8). Tables here stay in nominal $\mathrm{EIRP}_{5\sigma}$.

4.4. *Dwell-fraction completeness*

How long a carrier occupies one native channel defines the searched class, so we measure recovery against that directly. A carrier was injected into the retained per-integration spectra of 12 real windows, six coarse and six fine, at amplitudes of 2–20 times the per-integration noise and dwell fractions $f_{\mathrm{dwell}} = 0.10$–$1.00$, six realisations each: 3 888 trials, released with one row per injection (Table 3). Its detection criterion, recovered when the logged per-integration S/N reaches five, was re-derived from the released rows and agrees with the logged flag 3888 times in 3888 trials, with 0 mismatches across a clean separating gap. Recovery rises monotonically with dwell. Pooling amplitudes at or above 4σ, coarse windows recover 100 per cent of continuous carriers, 100 per cent at $f_{\mathrm{dwell}} = 0.25$ and 83 per cent at 0.10; fine windows give 100, 91 and 71 per cent.

An end-to-end check confirms this through the unmodified pipeline. Injecting a zero-drift, always-on tone into the calibrated visibilities of a coarse Band 7 window at 1, 3 and 10 times the per-integration rms and running the released extraction and search returns $T_\star = 20.5$, 65.3 and 217.7, each recovered within one channel of the injection and flagged as a detection. For 639 integrations the ideal coherent gain is $\sqrt{639} = 25$, so a carrier at the per-integration noise level should reach $\sim 25\sigma$: the measured 20.5 is that, less weighting and flagging losses.

Two distinctions apply. The dwell campaign injects into retained per-integration spectra, after the baseline step of Table 2, while the end-to-end check injects into visibilities, before it. One cell is also measured only in part, the through-pipeline recovery of near-static carriers on Class A windows, which no end-to-end zero-drift injection has yet constrained (Appendix B). The earlier claim that the pipeline suppresses persistent carriers is withdrawn.

The campaign measures completeness at the survey threshold and across configurations. The trigger acts on the stacked statistic and the de-drift step is an inverse-variance-weighted sum over integrations, so a carrier of per-integration amplitude A present for a dwell fraction f_{dwell} of a track of N_{int} integrations accumulates to

$$a \;=\; A\, f_{\mathrm{dwell}} \sqrt{N_{\mathrm{int}}} \tag{4}$$

in units of the per-integration noise. Recovery depends only on the combination a, so binning the released trials in a unifies the two axes and places all 12 injected windows on one axis. Those windows span $N_{\mathrm{int}} = 21$–$1\,314$, a factor of 63, over 11 targets in Bands 3, 6 and 7 and both resolution classes, so configuration-independence is measurable. In the 6 windows whose amplitude grid straddles the trigger, the 50 per cent recovery point lies at $a = 5.3$–8.3 (median 6.2), which is the 5σ threshold itself, independent of channelisation, integration count and band. In the other 6 the weakest injection already stacks to $a = 3.4$ and at least 97 per cent of every injection is recovered, consistent but uninformative about the curve's position. The pooled curves (Table 3) agree between classes within the sampling noise.

The injected carriers do not drift, so what is measured across 12 configurations is the persistent and partial-dwell, non-drifting class. On Class B windows that is the searched class, which therefore carries a survey-threshold completeness curve. On Class A windows the searched class is the drifting carrier, whose curve is still the single calibration configuration of Appendix B; the fine column of Table 3 constrains the near-static cell at the search stage only and bounds nothing about the transfer error. The quantity a is a derived combination, justified by the form of the statistic and by the collapse of 12 configurations onto one curve; it is not measured independently at every (A, f_{dwell}) pair.

One caveat attaches to the drift ceiling. The searched value is 12.0–$13.3\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$, line-of-sight $|a| = 3.60$–$4.00\,\mathrm{m\,s^{-2}}$, the generic value everywhere except the 12 windows where a known planet's own bound raises it. It clears Earth-rotation and Earth-orbit drifts by two orders of magnitude but is marginal against a close-in planet on a TRAPPIST-1 b-like orbit, so the limits of §6 are conditional on it (Figure 7).

Closure-phase vetting (Appendix C). This one-sided point-source test needs multiple independent baselines, so single-dish and beamformed surveys cannot perform it. It is implemented and simulation-validated, and at this survey's flux densities it has *no discriminating power*: per-baseline S/N never exceeds 5.1 in the positive-control fields, so no flagged window could have failed it whatever its nature. It is a proof-of-principle diagnostic and never an operative filter, and no window is described as having "survived" it.

Extended and cross-target screens (Appendices C and C). Satellite constellations grew across the 2013–2025 archival

TABLE 3
THE DWELL CAMPAIGN. *(a)* RECOVERY OF HIGH PER-INTEGRATION-S/N INJECTED CARRIERS AS A FUNCTION OF DWELL FRACTION, POOLING INJECTED AMPLITUDES $\geq 4\sigma$ PER INTEGRATION, FROM 3 888 TRIALS OVER 12 REAL WINDOWS. RECOVERY RISES WITH DWELL, A CONTINUOUSLY TRANSMITTING CARRIER BEING THE EASIEST CASE. THE EXPERIMENT ESTABLISHES THE DEPENDENCE ON TEMPORAL OCCUPANCY; ABSOLUTE COMPLETENESS AT THE SURVEY DETECTION THRESHOLD COMES FROM THE END-TO-END CAMPAIGN OF FIG. 6.

f_{dwell}	coarse windows	fine windows
1.00	100%	100%
0.50	100%	99%
0.25	100%	91%
0.10	83%	71%

(b) completeness at the trigger, against stacked amplitude a (Eq. 4), pooled over the 12 injected windows; trial counts in parentheses.

stacked a	Class B (coarse)	Class A (fine)
< 3	0% (30)	0% (72)
3–4	0% (12)	7% (42)
4–5	3% (30)	12% (42)
5–6	46% (24)	33% (48)
6–8	58% (60)	70% (96)
8–12	100% (108)	99% (174)
> 12	100% (1680)	100% (1470)

baseline. No known ordinary downlink allocation maps into the searched frequencies, but that excludes neither harmonics, intermodulation products nor unconventional emitters, and no exclusion claim rests on allocation alone. The primary defences are the control ensemble and the time-domain tests. A satellite enters star and control extractions alike within an integration, the channel-width mismatch and the drift-stacked statistic dilute transient passes, and a pass confined to one block and one target would evade the cross-target occupancy check by construction. A near-field emitter need not share a celestial source's far-field interferometric response, one reason a dedicated per-integration RFI screen and visibility-domain checks belong in any successor analysis. A chirped (quadratic-drift) extension, a per-channel periodicity search and a cross-target recurrence match at the same sky frequency all ran as demonstrations only. All three behave as the null predicts, none produces a candidate and no limit derives from them; Appendix C gives the rates, the pair counts and one bookkeeping caveat.

What follows is the final analysis. The pipeline as released differs from earlier working versions in five substantive ways, each logged with its effect on released products in Appendix I and the repository change-log; §5.4 dates the one that changes a released number. Asymmetric-derived quantities are historical there and enter no operative number.

Scope of the constraints. *What was searched:* 33 frequency islands totalling 93.1 GHz (88.2 GHz after molecular-line masking) at 15.3 kHz–31.25 MHz channelisation, during the analysed archival intervals, for carriers within the searched drift ceiling and one native channel in width, at nominal trigger thresholds whose measured recovery is 42 per cent (single configuration, transferred by assumption). Inside that box the non-detection is a constraint. *Nothing in this paper bounds the prevalence of:* (i) drift-rate structure on the 313 coarse windows, which carry no drift discrimination and no survey-threshold completeness curve (§4.4); (ii) transmitters emitting *outside the searched spectral windows*, meaning those 33 islands, not ALMA's frequency range (Fig. 2); (iii) transmitters active *outside the archival epochs*: the limits assume near-continuous emission and are only weakly constraining below epoch activity $\sim$0.1 (Table 11); (iv) drift rates *above* the searched ceiling $|a| = 3.60$–$4.00\,\mathrm{m\,s^{-2}}$, excluding the most close-in orbits (Fig. 7); (v) carriers placed *inside* the $\pm 50\,\mathrm{km\,s^{-1}}$ molecular-line mask, which is excluded search space and not searched-and-empty; (vi) emission *broader* than one native channel, for which the thresholds must be rescaled; (vii) anything below the nominal threshold, where measured recovery is 42^{+16}_{-14} per cent (Appendix B); (viii) any transmitter-frequency prior spanning ALMA's tuning range, under which the conditional bound vanishes (§6.2).

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5. RESULTS

The search, its non-detection and the per-window thresholds come first (§5.2, §5.3), then the four flagged windows and their dispositions (§5.4), and last the ancillary by-products, which gate nothing in the primary lane (§5.5). The non-detection is stated under two late-stage corrections, and both made the reported picture more conservative. The mid-analysis statistic revision (§5.4, Table 7) only ever removed flags (seven windows before, four after, none in which the star dominated its own controls), and withdrawing the coarse-window 0-of-500 recovery artefact (§4.3) restored the recovery of persistent near-static carriers rather than removing a claim.

Strict volume ordering (nearest first, §3) weights these results to the nearest stars, over 1.30–39.63 pc; per-target values, including each band's searched frequency range, are in the released catalogue (Data Availability).

TABLE 4
The survey as searched, and the paper's headline numbers in one place. Every count quoted in this paper reconciles with it; subset numbers are identified where used. The two completeness experiments measure different quantities and are kept in separate tables. The conditional searched-domain transmitter fraction is a methodological demonstration (Appendix J) and is deliberately absent.

Quantity	Value
Stars within 40 pc with qualifying public ALMA coverage	168
Stars searched; independent systems	88; 81
Star/band datasets	107
Spectral windows searched	431
extracted; repeats, defective, withheld	462; 21, 6, 4
Class A drift-resolved / Class B unresolved-excess	118 / 313 (73% Class B)
Execution blocks	102
Distance range	1.30–39.63 pc
Unique frequency union	93.1 GHz (33 intervals, 89.6–495.1 GHz)
Windows by ALMA band (3/4/5/6/7/8)	40/4/4/216/155/12
Effective after line-exclusion mask	≈88.2 GHz
Native channel widths	15.3 kHz–31.25 MHz; median 15.625 MHz
Drift-rate ceilings	1.1–5.9 $\mathrm{kHz\,s^{-1}}$ (12.0–13.3 $\mathrm{Hz\,s^{-1}\,GHz^{-1}}$)
On-source time per window	21–5274 s (median 2087 s)
Nominal $S_{\min}$	0.48 mJy–1.30 Jy (median 6.7 mJy)
Nominal $P_{\rm trig}$, Class A (15.3–1953 kHz)	2.3×10^{13}–6.8×10^{16} W (median 7.5×10^{14})
Nominal $P_{\rm trig}$, Class B (15.625–31.25 MHz)	1.6×10^{13}–2.8×10^{17} W (median 2.0×10^{15})
worst-case Hanning-corrected medians	2.0×10^{15} / 5.4×10^{15} W
Completeness, by class and experiment	Table 8 (amplitude); Table 3 (dwell)
Threshold crossings ($\geq 5\sigma$ at the stellar position)	20 windows (all Class A)
Star-exceeds-ring windows ($\geq 5\sigma$)	4 windows (all Class A)
circumstellar or foreground CO	3
unattributed, retired on non-recurrence	1
Credible technosignatures	0

5.1. *Barnard's Star and Wolf 359*

Barnard's Star and Wolf 359, the second- and fourth-nearest stellar systems, have not to our knowledge been searched for technosignatures at ALMA frequencies before. Both are non-detections in public Band 6 archival data: nominal $\mathrm{EIRP}_{5\sigma}$ trigger thresholds of 6.2–6.9×10^{13} W (Barnard's Star, 4 windows) and 7.8–9.1×10^{13} W (Wolf 359), with continuum upper limits of 0.575 and 0.453 mJy.

5.2. *Data-quality exclusions and open items*

107 target/bands are searched to completion and reported here. Four classes of entry are absent from the released catalogue and itemised in Appendix I: four process-level search failures, each still carrying a continuum non-detection; one pair excluded for data quality; windows withheld for a noise-handling defect; one relabelled entry.

A further 68 candidate target/bands were selected but never delivered a searchable product, and the cause lies in the selection step. Re-running the crossmatch against the archive shows that 60 of them were never observed at all, since no observing unit offered to those entries contains the star. They sit a median 1.6 degrees from the nearest field centre, because the observations were pointed at solar-system targets whose quoted field of view is enormous, and a candidate crossmatch that trusts that quantity will admit stars far outside any real primary beam. The per-star pointing ledger is released. The remaining 8 entries are genuine processing failures; re-running the failed set on the current code returned 38 further failures in 39 attempts, its one recovery falling outside Bands 3 to 8 and outside this release. A positional crossmatch must not use the quoted field of view of an ephemeris or solar delivery as a search radius. 53 of the 168 stars admitted to the candidate list fail on pointing in every band, so the sample here is 88 of about 115 genuinely covered stars rather than 88 of 168, and the coverage completeness of Table 4 should be read against the smaller denominator.

THE COARSE-NOISE DEFECT, AND THE BOUND ON ITS REACH

A reproducible defect produces implausibly small noise in a minority of coarse-channel windows. The radiometer relation $\sigma \simeq \mathrm{SEFD}/\sqrt{N_{\rm bl}\Delta\nu_{\rm ch} t_{\rm on}}$, in the system equivalent flux density (SEFD), makes $q \equiv \sigma\sqrt{t_{\rm on}\Delta\nu_{\rm ch}} \simeq \mathrm{SEFD}/\sqrt{N_{\rm bl}}$ a quantity physics predicts, so a window whose q lies far below the sample median is too extreme for a noise fluctuation and points instead to a bookkeeping defect; we exclude such windows at 100× below the median. Six of 431 windows fail, across five stars, and their nominal thresholds of 2.0–7.8×10^{12} W would otherwise have set this paper's headline depth; excluding them, the best is 1.6×10^{13} W.

The retained population is itself broad, so the exclusion does not rest on its narrowness; q runs over a factor ∼290 of the sample median, mostly the band dependence the same relation predicts through SEFD. It rests on the size of the gap, the six sitting a factor 1.5×10^{3} below the lowest retained window across an empty 3.2 decades, so that any threshold between 1.5×10^{1} and 2.2×10^{4} in absolute q units selects the same six. It rests also on direction: σ is estimated from the control probes and applied to every position of a window alike, so a window-level deflation can only over-state sensitivity, never manufacture a star-exceeds-ring window. All six have on-source times of 12 to 48 s, and no integration-time cut is applied anywhere.

The six share a signature rather than an origin in the observing programme: all are 12 m array, 3 in Band 6 and 3 in

TABLE 5
The four windows containing a stage-1 spatial outlier, with the evidence dispositioning each. $T_\star$ is the search statistic at the stellar position (Eq. 2) and 'ring max' the largest of that window's 512 control statistics. $\nu_{\rm cross}$ is the crossing channel, which differs from the window centre by up to 0.85 GHz here, and $S_{\rm peak} = T_\star\sigma$ is the peak flux density in one native channel. $\Delta\nu$ and Δv are the offset of that channel from the nearest *laboratory* rest frequency of the masked species, topocentric, as $c\,\Delta\nu/\nu_{\rm rest}$ and so positive when the observed frequency exceeds the rest frequency; the $\pm 50\,{\rm km\,s^{-1}}$ mask applies in the stellar frame after the conversion chain of Appendix A, and the repaired mask of §5.3 changes no row. The frozen mask stored rest frequencies rounded to 1 MHz, so the released offset column reads 0.202 MHz ($0.53\,{\rm km\,s^{-1}}$) smaller than the values here for CO(1→0). No disposition rests on a single test. The HD 48370 row is the limiting case, a near-tie of rank dispositioned by velocity coincidence with independently published emission (§5.3).

Window	$\nu_{\rm cross}$ (GHz)	$T_\star$	ring max	$S_{\rm peak}$ (mJy)	$\Delta\nu$; Δv (MHz; ${\rm km\,s^{-1}}$)	Disposition
β Pic B3	115.260644	14.65	7.27	54	-10.56; -27.5 (CO 1→0)	circumstellar CO
β Pic B6	230.516128	11.68	9.12	78	-21.87; -28.4 (CO 2→1)	circumstellar CO
HD 48370 B6	230.511674	27.10	26.83	0.82×10^3	-26.33; -34.2 (CO 2→1)	foreground CO
CP−72 2713 B7	344.269746	5.81	5.68	12	-1526.24; -1323.2 (CO 3→2)	unattributed; no recurrence

Band 7, all at 15.6 MHz, in 5 execution blocks across 4 projects, and every one has a normally behaved finer-channelised window covering the same frequencies in the same block (Appendix I), which is what an auxiliary window riding on a science baseband looks like. Modelling q for every retained window from its own band, array and antenna count leaves those residuals unimodal with the six 3.0 dex below the lowest retained window and nothing between, so no continuum of smaller deflations hides under the cut.

21 of the 462 extracted rows are duplicate catalogue entries for one physical window, the same (star, execution block, frequency range, channel width) key appearing twice in the archive query, either as repeated archive products or as alternative calibrations of one execution block, so one row per key is the right statistical unit. The retention rule is deterministic and quantity-independent, the key's first-listed row in the archive query's own identifier order, so it consults no measured value and cannot prefer a crossing or a deeper threshold. Repeats need not agree; two reductions of the same τ Ceti Band 6 window differ by 44 per cent in threshold, which exceeds the flux-scale systematic of §4.2 and is the most direct empirical bound here on threshold reproducibility. The two HR 1010 Band 6 rows straddle 5σ at 5.10 and 4.95, so the crossing count is rule-dependent at the level of one window. The four flagged windows are not, no discarded row being both a crossing and star-exceeds-ring (Appendix I).

5.3. *Technosignature search*

All 107 star/band datasets yielded at least one carrier-lane result (process-level failures, §5.2, uncounted), spanning 431 spectral-window measurements after the catalogue audit removed 21 duplicate rows (Appendix D); B6/216 and B7/155 dominate (per-band breakdown in Table 10). ${\rm EIRP}_{5\sigma}$ trigger thresholds span 1.6×10^{13}–2.8×10^{17} W, median 1.4×10^{15} W (Figures 2, 1); the deepest thresholds are reached toward the nearest systems (1.6×10^{13} W for the UV Ceti pair at 2.7 pc), though depth also tracks integration time and channelisation, and the shallowest is η Corvi. Against the benchmarks of §4.2, only that pair's two resolved components reach below the Arecibo-like planetary-radar EIRP, 1 system of 81.

The statistics come strictly before the astrophysics, in three named stages that do not promote one another. A *crossing* is one channel×drift cell at the stellar position reaching $T \geq 5$: 20 of the 431 windows contain one. A *stage-1 spatial outlier* is a window whose stellar statistic also exceeds all 512 controls: four do. That screening device selects a window for examination (§4.1). Each of the four is then carried past that floor by a second-stage local null ((v) below), which asks whether a feature exceeds the noise at its own position without discriminating spatially; in 3 of the 4 windows the control-ring maximum clears it too, so candidacy still turns on the first-stage star-versus-control comparison. A *candidate* is a stage-1 flag surviving every veto, and there are none. Three of the four flags have independent CO evidence, matched transitions, velocities and epochs at β Pictoris and published foreground emission at HD 48370; the fourth, towards CP−72 2713, never acquired one and is retired on a second archival epoch, below. What disposes of the three is that astrophysical evidence, with the $\pm 50\,{\rm km\,s^{-1}}$ mask playing no part. Table 5 gives each flagged window with the evidence dispositioning it.

None of the 4 stage-1 flags exceeds this survey's own calibration or separates from the empirical background and known astrophysical emission.

SURVEY-LEVEL FALSE-ALARM CALIBRATION

Each window-level test compares the peak signal-to-noise ratio at the star's position with the same statistic at 512 controls. The drift grid and thousands of channels apply identically to both, so the per-window trials factor is absorbed empirically. The statistic is symmetric, one propagated stellar position against one position per control. Every number here derives from that frozen statistic; the retired asymmetric variant is development history (Appendix I). The false-alarm probability has five components, in the order the procedure applies them.

(i) The empirical spatial rank probability. The per-window significance quantity is an empirical spatial rank probability under the control-position null, and no Gaussian-tail p-value; what the control ensemble delivers is an empirical null distribution conditional on approximate spatial exchangeability, and not a direct measurement of a local false-

alarm probability. Under exchangeability one designated position of 513 ranks first with probability exactly 1/513, whether the noise is Gaussian or heavy-tailed, so heavy tails cannot explain an excess. The survey expectation is $431/513 = 0.84$ windows, and $P(\geq 5\,|\,0.84) = 1.7 \times 10^{-3}$ under an independent-Poisson approximation. That excess is astrophysical: in 3 of the 5 rank-first windows genuine celestial line emission sits at the stellar position (Table 5), so they are not draws from the noise null, and removing them leaves 2 against 0.84 expected ($P = 0.21$).

(ii) The $\geq 5\sigma$ amplitude requirement and (iii) the line mask. Only 20 of 431 windows contain any $\geq 5\sigma$ crossing at the stellar position. The ensemble calibrates the rank and the trial count follows from it, because a window's channel×drift grid holds up to 6 866 208 cells (Appendix H), far more than 512 controls can resolve, so the control maximum absorbs whatever effective trial count the noise carries. The velocity-space mask (Appendix E) then removes known molecular emission from every crossing before it can count as unattributed. It is an analysis choice, never lowers the noise-null budget, and carries no disposition alone, the three CO attributions being kinematic (Table 5).

(iv) The family-level pseudo-target Monte Carlo. Each trial draws one control per window uniformly from its annulus as the pseudo-star and passes it through the operative gates, $T \geq 5\sigma$ then the velocity mask. This prespecified procedure carries threshold, mask and ring geometry together on real data and expects 0.20 flagged pseudo-target windows (82 per cent of trials flag none); (i) and the Bonferroni scale agree.

(v) The control-ensemble floor, and the second-stage local null. Authentication falls to localisation, recurrence and astrophysical attribution (§4.1), and we pre-commit that no crossing will be advanced as a candidate on rank statistics alone. That floor belongs to the first stage, and a second-stage local null carries the four flagged windows past it. For each we take a retained control position, standardise its residual by the pipeline's per-integration noise map, shift every integration independently and at random in channel, and re-run the identical search, with the same noise map, weights, drift grid and channel count. A rigid shift preserves each integration's spectral autocorrelation, 0.65 at lag one, and the maximum runs over the same grid, so each null realisation carries the same trials factor as the measurement at the star. Independence across integrations leaves nothing coherent along a drift track, so each realisation is signal-free by construction, and between 2×10^4 and 10^5 realisations per window resolve probabilities well below 1/513. The statistic was reimplemented from the pipeline source and reproduced all four published $T_\star$ values bit for bit before any null was drawn.

Each null has first to reproduce the distribution of the 512 real control maxima, genuine draws of the same statistic at exchangeable positions. The criteria were fixed before inspection: 0.2 to 5 per cent of real controls above the null's 99th percentile, under 1 per cent above its maximum. Three windows pass. HD 48370 Band 6 fails badly: 375 of its 512 real controls, 73 per cent, exceed the maximum of 10^5 null draws, and those controls sit at median 9.30 against the null's 4.72. The cause is physical, the field carrying extended emission and its imaging residuals, coherent across integrations, which the scramble destroys by design. A scrambled-noise probability there would be manufactured significance, so we quote none for that window.

The two β Pictoris windows act as positive controls: no draw reaches $T_\star$ in either, giving $p < 1 \times 10^{-5}$ in Band 3 and $p < 5 \times 10^{-5}$ in Band 6, which is what a working statistic should return where a genuine localised astrophysical signal is present, here on independently known CO. For CP−72 2713 the null resolves the value itself, 157 of 10^5 draws equalling or exceeding $T_\star$, and an extreme-value fit to that window's 512 control maxima agrees with it to a factor 1.3, a 2000-fold bootstrap putting the chance of the true value falling below the Bonferroni scale at 1.3 per cent; the values, and what they do and do not license, are with that window below. The same fit places HD 48370 at $p = 1.5 \times 10^{-2}$ (8.3×10^{-3} to 2.2×10^{-2}), unremarkable and consistent with the CO attribution.

Two limits attach to these numbers. The retained products keep full dynamic spectra for 8 of the 512 control positions, so position-to-position heterogeneity rests on 8 draws; that is what breaks the HD 48370 null, a limit the retained products impose on the method and one that retaining more control spectra would remove. Parametric extrapolation of a null tail is not used for any number quoted here: for β Pictoris Band 3 the natural fits disagree by 26 orders of magnitude and one implies a bounded tail and returns exactly zero.

Exchangeability is the central assumption of the statistic: the rank expectation holds only if the star and its 512 controls are exchangeable under the null, which spatially varying noise, sidelobes, correlated pixels or phase-centre-localised calibration residuals could spoil. For a debris-disc archive the dominant violation is none of those but resolved astrophysical emission filling the annulus, measured here in HD 48370's CO ring and in β Pictoris' recurring Band 6 window, where the ring outshines the star by 1.5×. It suppresses a star-exceeds-ring outcome and never manufactures one, so it costs sensitivity alone. The primary-beam term is settled by construction (§4.1): the statistic is formed on *uncorrected* amplitudes, whose thermal noise is position-independent across the field, and $\sigma(t, \nu)$ is one scale shared by the star and every control. The 513 positions are therefore exchangeable under the noise-only null by construction, and the primary beam enters only in converting to flux. The rest we measure by stratum (Fig. 3). Median per-window control maxima are flat across bands, 4.46, 4.59, 4.42, 4.52, 4.63 and 4.69 in Bands 3 to 8, and a Kolmogorov–Smirnov test of the star's rank among its controls against $U(0, 1)$ is consistent in every band ($p = 0.20$–0.83) and in the coarse stratum ($p = 0.65$). Fine-channel windows sit higher (control maximum 5.77 against 4.41), as their far larger channel×drift trial count predicts; the test is conditional on the window, so that is no violation.

For the stronger question, exchangeability of the ensemble itself, we ranked each of a window's 512 controls against the other 512-1 by the statistic and add-one rule used for the star, pooled over the survey. The 220 672 pseudo-star ranks are uniform to the resolution the ensemble can express ($p = 0.37$) in every band and at both channelisations; the mean of 0.5010 is the construction value under the add-one rule and carries no information. That test compares a control against the other 512-1 plus add-one, a different geometry from star-against-ring, so it cannot by itself detect a centre-versus-annulus asymmetry. **The test that can is the two-sample comparison of the 431 real stellar**

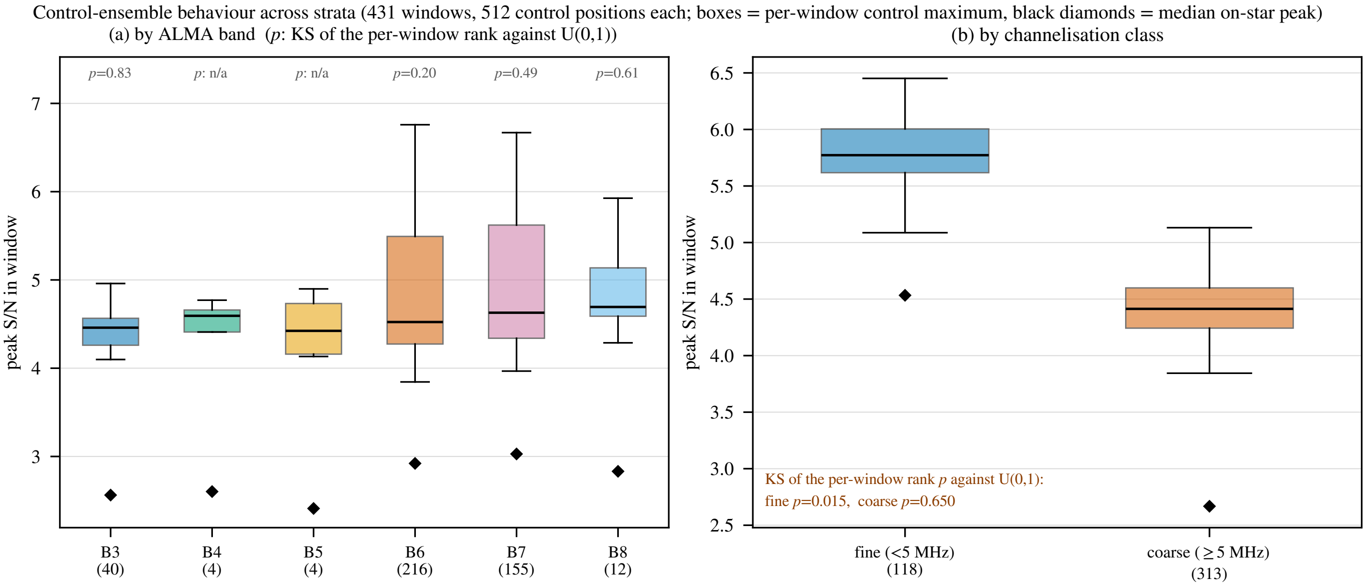

FIG. 3.— Exchangeability of the 512-position control ensemble by stratum. Boxes: per-window control maximum; diamonds: median on-star peak; p: the Kolmogorov–Smirnov test of the text. The ensemble tracks the on-star statistic in every band and both channelisation classes. The one departure is the drift-search class ($p = 0.015$), driven by the four flagged windows of Table 5.

ranks against the pooled pseudo-ranks, $p = 0.38$: the stellar position behaves like one more position in its own annulus, their median *statistics* agreeing at 2.89 against 2.89 ($D = 0.030$). The radial test is the second: the Spearman correlation of control statistic against reconstructed radius has median 0.009 over the 304 crossing-free windows, with 5th and 95th percentiles -0.08 and 0.11, so any systematic radial suppression of the controls is bounded below 11 per cent in the mean. Both are null, which supplies the bound the one-sidedness argument above lacks.

One stratum departs, and we report it separately. The fine-channel windows fail the same $U(0,1)$ test at $p = 0.015$ ($n = 118$): four contain an on-star peak above their own control maximum, against 0.22 expected, precisely the four flagged windows of Table 5. That fits the astrophysical line features identified below, with no misbehaving ensemble, but the departure is genuine, so fine-window rank probabilities are slightly optimistic. It is also localised, since excluding the 20 fine windows with any $\geq 5\sigma$ stellar crossing restores uniformity ($p = 0.39$, $n = 98$). Per-control noise records and gain are two stratifications the frozen products cannot support, so the empirical rank distributions bound the net effect: expected flags 0.20 against a rank-only budget of 0.84, with no excess in any stratum. The windows nearest the beam edge are not silently affected either: the largest retained offset, j1256-1257 at $0.46\,\theta_{\mathrm{PB}}$, sits inside the annulus's own radial span, and the four ϵ Eri Band 6 windows are withheld outright (§4.2). Controls at equal holography-model gain would remove the sky-brightness term.

THE FULLY SYMMETRIC REGION-MAX CHECK

A fully symmetric region-max variant, applying identical source-region, drift-rate and frequency maximisation at the star and at every control, needs products the frozen release does not carry, so its domain is seven locally reprocessed windows. All give $T_\star = 3.7$–4.9 against control maxima of 6.7–7.7, $p = 0.81$–1.00, so the star is exchangeable with its ring wherever the test runs at full symmetry. The single-position calibration above remains the operative release-wide criterion; the two are never interchanged (Appendix H).

ASTROPHYSICAL END-TO-END VALIDATION: β PICTORIS CO

Two of the four flagged windows are towards β Pictoris, a debris disc with resolved CO emission (Matrà et al. 2017; Dent et al. 2014): one in Band 3 and one in Band 6 (Table 5), each offset from its CO rest frequency by more than the automated one-channel check allowed, which is why that check missed both. Through the frame chain (Table 6) the offsets are one kinematic component, +0.44 and -0.29 km s^{-1} from stellar systemic, in the two lowest CO rotational transitions of a system with detected circumstellar CO. The unflagged Class A window at 230.528134 GHz, a third epoch six years earlier, independently shows CO at -0.46 km s^{-1}: all three epochs and both transitions lie within 0.46 km s^{-1} of systemic, deep inside the $\pm$50 km s^{-1} mask. The stage-1 Band 6 window carries 114 formal $\geq 5\sigma$ crossings where a channel-confined carrier admits one; the retained products do not record whether those channels are contiguous, so the count bounds the feature's extent at 2.26 km s^{-1} without measuring it (Appendix C for the tabulation cap). Peak flux densities of 54 and 78 mJy per channel, ratio 1.4, are the right order for this belt's spatially integrated CO though below the optically thin LTE ratio ($\simeq 4$). Excitation cannot account for a deficit that size, since Matrà et al. (2017) find this gas only mildly subthermal (CO 3−2/2−1 $= 1.9 \pm 0.3$ against 2.25 in LTE); beam dilution and resolved-out flux can, the synthesised beams being 2.71 and 1.09 arcsec; and since both channels are far narrower than the line, the peak per channel is comparable across the two despite their differing widths.

We attribute both crossings to this kinematically offset CO. Within the frozen release the validation rests on the velocity chain of Table 6, which recovers the whole check list untuned: frequencies to the frame chain's stated tolerance, velocity concordance across epochs and transitions, localisation against the control ensemble, and drift class zero to within the grid.

Attributing the pair to CO does not diminish it. Measured on the operative symmetric statistic and against the survey background, the Band 3 peak $T_\star = 14.65$ is matched or exceeded by 5 of the 431 per-window control maxima (add-one $p = 0.014$ on a denominator of 432) and the Band 6 peak $T_\star = 11.68$ by the same 5 (add-one $p = 0.014$). Those 5 include this star's own CO-filled Band 6 ensemble at 15.2, the 5th largest control maximum in the survey, below HD 48370's CO(2→1) ring at 26.8 and its ^{13}CO ring at 20.4. The alternative needs a transmitter coincidentally placed at matched offsets from two CO transitions in the same star. Every flagged window, and the crossing the statistic revision removed, belongs to one population: β Pic, AU Mic and CP−72 2713 are β Pictoris moving-group members with debris discs, and HD 48370 is a disc host behind a molecular cloud. The searched sample is, to a good approximation, the ALMA debris-disc and young-moving-group archive, the worst case for astrophysical false positives at millimetre wavelengths, so the flags concentrate where the foreground is richest.

The same population raises the stellar-emission question directly, and the search suppresses it twice. Gyrosynchrotron and free-free emission from chromospheres and coronae are broadband, while the block-median baseline of §4.1 removes everything broader than its 65-channel passband, ∼32 MHz at the fine class, a fractional bandwidth of $\sim 10^{-4}$; MacGregor et al. (2020)'s AU Mic flares are coherent across the full 8 GHz. Flares are also time-confined, so the inverse-variance stack over all integrations dilutes them. What survives is the multiplicative route of §4.1, a flare continuum times a spectral calibration residual, bounded there and measured for CP−72 2713 below. The search is blind to the time axis by construction, and the retained dynamic spectra are the only handle on it.

DISPOSITIONS OF THE REMAINING FLAGGED WINDOWS

The third, towards HD 48370 in Band 6, is the largest excess here in absolute and the smallest in relative terms: $T_\star = 27.10$ against a control-ring maximum of 26.83, the star exceeding its ring by one per cent of the statistic. Star and ring rising together is what emission filling the primary beam looks like. The window sits $-34.2\,\mathrm{km\,s^{-1}}$ topocentric from CO(J=2→1). The attribution is independently published: Cataldi et al. (2023) report prominent CO(2→1) and ^{13}CO(2→1) emission towards HD 48370 at $v_\mathrm{bary} \approx 41\,\mathrm{km\,s^{-1}}$, well separated from its $23.6\,\mathrm{km\,s^{-1}}$ systemic velocity and noted by them as potential foreground cloud contamination; our frame chain returns $v_\mathrm{bary} = +41.2\,\mathrm{km\,s^{-1}}$, so we recover it and confirm it kinematically. Two further checks close the alternative. The same authors report the HD 48370 disc undetected in both CO and carbon, so none of this emission can be circumstellar; and the sight line runs through the outer Galaxy at $l = 214.55°$, $b = -3.19°$, where a flat rotation curve predicts $V_\mathrm{LSR} = +23.1\,\mathrm{km\,s^{-1}}$ for gas a couple of kpc away, against the $+23.9\,\mathrm{km\,s^{-1}}$ our own LSR chain measures. That is what Galactic rotation gives for a foreground cloud in that direction and nothing local could produce it. The peak is 0.82 Jy in one 122-kHz channel with the ring at the same level, which is a bright line filling the beam. The ^{13}CO(2→1) window of the same execution block has a control maximum of 20.4 against only 4.6 at the star, so there is bright ^{13}CO in the field but not at the star, as extended foreground emission predicts. The reconstructed annulus geometry puts that peak 9.2 arcsec from the star, which against the window's 5.55-arcsec synthesised beam is 1.7 beams: the emission is resolved away from the star and clustered off-centre, with the per-control radii in the released catalogue. The Band 8 tuning says the same thing by omission: all twelve Band 8 windows here belong to η Crv, HD 48370 and HD 61005 and are [C I] $^3P_1$–$^3P_0$ (492.160651 GHz) tunings, and HD 48370's carries a 5.05σ on-star crossing against a ring maximum of 6.23, which is not one of the four stage-1 flags because its own ring exceeds it. Its crossing channel sits -129 km s^{-1} from [C I], so it is not [C I] emission from the disc, the cloud or the star; its own ring dispositions it as an ordinary noise crossing. What the window does show is no line at the star in this disc's carbon tuning, which is what Cataldi et al. (2023) report. The rank statistic identifies nothing here, a stated limitation of the spatial test.

CP−72 2713: AN UNATTRIBUTED FLAG, AND ITS RETIREMENT

The fourth flag, towards CP−72 2713 in Band 7, never acquired an astrophysical attribution and was marginal throughout. Its prior for one is high, the star being a K7/M0 member of the ∼24 Myr β Pictoris moving group with a cold, dust-rich debris disc detected in ALMA 1.33-mm continuum (Moór et al. 2020), the same population and class of magnetically active star as AU Mic. We recorded the window as an unattributed stage-1 flag, a statistical or astrophysical feature rather than candidate evidence, under criteria pinned in advance: *promote* on independent confirmation, *retire* on a clean second epoch or on identification with catalogued emission. That wording is committed to the released repository on 2026 September 11 (`119a68a6ab50`), 9 hours before the second block was searched.

That same member observing unit set, `uid://A001/X2d20/X2e25`, holds a second public execution block at this tuning, `A002_Xff0235_X502d`, which the per-target download budget of §3 declined. We have since calibrated and searched it with the unmodified pipeline. It matches the first epoch in tuning, channel width (488.28 kHz), channel count (3534), integrations (495) and on-source time, and the two frequency axes are registered to 87.6 kHz, 0.18 of a channel, so the feature falls in the same channel, 35. Only the drift grid differs, one trial coarser, which puts the first epoch's fitted rate 0.12 channel from the nearest node. The second block is also the deeper, combined rms 1.964 mJy against 2.097, so a persistent emitter at the first epoch's flux would have appeared there at $T_\star = 6.21$, above both the 5σ trigger and that window's largest control, 5.93. That margin is narrow, so a second *flag* was never assured; what the retirement rests on is the flux comparison below, which carries no trials factor.

The feature does not recur. The first-epoch feature drifts. Its maximum lies at a drift of $-2.06\,\mathrm{kHz\,s^{-1}}$, a track sweeping 15.8 channels across the block, so the test belongs at that frequency *and* that drift. There the second epoch measures -1.0 ± 1.8 mJy, -0.6σ, against 11.3 ± 2.0 mJy in the first. Read as two measurements of one constant amplitude, the pair is inconsistent at 4.6σ; with the first epoch's flux taken as exact, which flatters the test because that flux is the largest value in the window and so biased high, the second epoch alone excludes it at 6.8σ. Maximised afresh over the second epoch's own drift grid, the same channel returns $T_\star = 3.60$ against 5.81, worth 1.6σ with an arbitrary recovered rate, and that weaker comparison is the form in which such tests are usually quoted. Two epochs cannot separate a noise excursion from an intermittent emitter, and we do not claim they do.

Nothing else in the block is elevated. At the feature channel $T_\star = 3.60$ stands 0.6σ above the ten neighbouring channels (mean 3.23, standard deviation 0.63), the noise-only expectation for a maximum over drift trials. The window's largest excursion, 4.70, ranks 188 of 513 against the same 512 controls the first epoch topped, add-one $p = 0.37$, and the largest control reaches 5.93, so the annulus alone now throws an excursion bigger than the 5.81 that raised the flag.

One qualification travels with this result. The two blocks are a single night's pair, 2022 October 2, beginning 00:40 and 02:42 UT with a 60-minute gap between them, so the second measurement is 2.05 h after the first and the pair spans 3.1 h. That makes the test strong against the class the scramble null below cannot reach, because a correlator spur, a fixed-channel bandpass residual or a birdie is at its most reproducible two hours later in the same configuration and tuning, and none reproduced. It also constrains persistence across two hours alone, leaving the duty cycles of Appendix J untouched. AU Mic's two epochs (§5.4) are ten weeks apart, a different test.

The second-stage local null replaces the 1/513 rank resolution with a measured probability, $p = 1.6 \times 10^{-3}$ from 10^5 scramble draws on a control and 2.0×10^{-3} on the star's own residual, with $p = 1.2 \times 10^{-3}$ from an extreme-value fit to this window's 512 control maxima, every one of them above the Bonferroni scale. The scramble shifts every integration independently in channel, destroying anything coherent across integrations and with it any frequency-stationary instrumental feature: a correlator spur, a fixed-channel bandpass residual, a birdie. A small local-null probability disfavours thermal noise alone and is not evidence against an instrumental origin.

Evidence that framed the flag survives as supporting detail. The feature is at 344.269746 GHz, not the window centre 345.1152 GHz quoted in earlier versions, and channel 35 of 3534 makes it the most edge-proximate crossing in the survey, 0.010 of the way into its own window. $T_\star$ exceeds the ring maximum by 0.13, the smallest margin 512 controls can express, so the margin records a resolution floor alone.

Frequency disposes of the catalogue entries. At zero drift the first-epoch statistic here is only 2.02σ, while a catalogued transition is stationary in the observed frame to well within a channel over two hours, so none can produce this feature. Re-querying the wider catalogues at the corrected 344.269746 GHz, filtered to species observed in space and pushed through this paper's frame chain (Appendix A), returns 3 entries in the $\pm 50\,\mathrm{km\,s^{-1}}$ tube in both the stellar and the LSR frame, of which SO(8_8–7_7) at 344.310612 GHz is no obscure one: it is absent from the 17-transition mask of §5.3, and had it been present the feature would have been masked at -12.3 $\mathrm{km\,s^{-1}}$ in the stellar frame. Width does not contradict that reading, the feature occupying one 488 kHz channel against the $\sim 4\,\mathrm{km\,s^{-1}}$ of a belt at 140 au. The same query at each of the 20 crossing frequencies, counted topocentrically, returns at least one catalogued transition within 60 $\mathrm{km\,s^{-1}}$ for 19 of them, median 3, so bare catalogue proximity has no discriminating power at these frequencies and the earlier reliance on it was misplaced.

The star's debris belt, resolved by Moór et al. (2020) at 140 au (3.8 arcsec at 36.7 pc), lies six beams from the flagged window's 0.61-arcsec extraction, so the disc is not the feature; the corresponding CO(2→1) mass limit comes from that star's Band 6 window and is released with the catalogue. A split-half of the retained 495 integrations gives $T_\star = 3.08$ and 5.12 against 4.11 expected for a persistent source, so both halves carry the feature and an event confined to either one is excluded; the 1.4σ difference between them is consistency, not exclusion. The per-integration continuum of 0.07 ± 0.06 mJy would need $\epsilon \approx 70$ to drive the multiplicative bandpass route of §4.1. Polarisation hands, baseline splits, visibility-domain localisation and calibrator comparison need re-extraction beneath the frozen products, and we claim no test we have not run.

What carries the retirement is the matched-frequency, matched-drift null of the second epoch, with the local null that never placed the feature near the Bonferroni scale and the trials budget of the survey behind it. The pinned criterion is met, so the flag is retired as a candidate on the balance of two independent measurements.

A non-recurrence is only meaningful if the same machinery can recover a repeat when one exists. We therefore applied it to emission we already believe in. Bookkeeping first, since blocks and epochs are not the same thing: an observing unit holds several execution blocks, the download budget of §3 took one per unit, and blocks of one unit are not independent epochs. The unit behind the Band 3 flag holds 4 blocks and all 4 are now searched unmodified, an exhaustive control. The CO($1 \to 0$) feature is present in every one, at $T_\star = 14.65, 17.29, 27.68, 30.76$, with 17, 24, 24, 24 channels above threshold and the control maximum below the star throughout (largest 7.34 against a smallest $T_\star$ of 14.65). Those executions share one scheduling block within 23 h and are one epoch repeated. A second Band 6 CO($2 \to 1$) window, 230.528134 GHz in a different and also exhaustive unit (2 of 2, 25 d apart), crosses the threshold without becoming a stage-1 spatial outlier: $T_\star = 10.16, 9.95$ on an exact configuration match (244.14 kHz, 3770 channels), integrated line flux 984 ± 88 against 911 ± 69 mJy MHz, differing by -0.7σ. It is Class A at 244.14 kHz, so the recurrence is demonstrated at the survey's finest drift discrimination. The stage-1 Band 6 window at 230.516128 GHz is out of this test for a different reason: its unit holds a second block, A002_Xd9668b_Xa9df at 48.9 GB, which has not been searched; no second-epoch measurement of that window exists either way.

The recurring window also locates the emission. In both blocks the annulus is brighter than the star (ring maxima 15.17, 15.08 against $T_\star = 10.16, 9.95$, 8 and 6 of the 512 controls above the star), and that annulus, 3.82–21.27 arcsec,

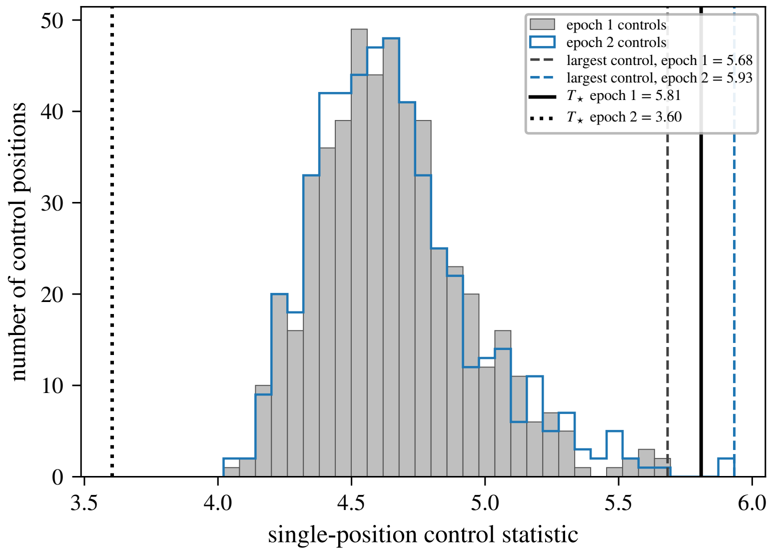

FIG. 4.— The CP−72 2713 Band 7 crossing inside its own control ensemble, in both epochs. Histograms give the 512 single-position control statistics of that window in epoch 1 (filled) and in the second block 2.05 h later (outline); the two agree, which is the evidence that the datasets are comparable. Vertical lines mark $T_\star$ and the largest control in each. In epoch 2 the control annulus alone reaches 5.93, beyond the 5.81 that raised the flag, while the star falls to 3.60.

TABLE 6
LINE-ATTRIBUTION AUDIT OF THE THREE LINE-ATTRIBUTED WINDOWS (FIRST THREE ROWS) AND THE CLASS A β PICTORIS BAND 6 WINDOW AT 230.528134 GHZ, WHICH CARRIES NO STAGE-1 FLAG YET SHOWS THE SAME CO COMPONENT, THROUGH THE FRAME CHAIN OF §4.2. TOPOCENTRIC OFFSETS ARE OBSERVED MINUS CATALOGUED REST FREQUENCY AS $c\,\Delta\nu/\nu_{\rm rest}$ (CO J=1→0 115.271202 GHZ, CO J=2→1 230.538000 GHZ, PICKETT ET AL. 1998; THE FROZEN MASK'S OWN ENTRIES, ROUNDED TO 1 MHZ, ARE NOT USED HERE), SO A NEGATIVE OFFSET MEANS THE OBSERVED FREQUENCY LIES BELOW REST; THE BARYCENTRIC CORRECTION IS THE EARTH-EPHEMERIS TERM AT THE EXECUTION-BLOCK MID-TIME (BLOCK UIDS IN THE MACHINE-READABLE RELEASE). THE AU MIC SYSTEMIC VELOCITY IS CATALOGUED (SIMBAD, *Gaia* DR2, FOUQUÉ ET AL. 2018). FOR β PICTORIS WE ADOPT THE GAS-DERIVED $v_{\rm sys} = +20.0 \pm 0.7$ KM S^{-1} OF MATRÀ ET AL. (2017) RATHER THAN THE *Gaia* DR3 CATALOGUE VALUE OF $+16.84$ KM S^{-1} (GAIA COLLABORATION ET AL. 2022): THE STAR IS A6V WITH $v \sin i \simeq 130$ KM S^{-1}, INSIDE THE HOT-STAR TEMPLATE REGIME IN WHICH DR3 RADIAL VELOCITIES CARRY A KNOWN SYSTEMATIC (BLOMME ET AL. 2023).

Flag	UTC start	$f_{\rm obs}$ (GHz)	Transition	Topo. offset (MHz; km s^{-1})	Bary. corr. (km s^{-1})	$v_{\rm sys}$ (km s^{-1})	Stellar-frame offset (km s^{-1})
β Pic B3	2022-03-03 21:49	115.260644	CO(1→0)	-10.56; −27.46	−7.90	+20.0	+0.44
β Pic B6	2019-03-12 00:14	230.516128	CO(2→1)	-21.87; −28.44	−8.15	+20.0	-0.29
AU Mic B6	2014-08-18 04:01	230.825230	CO(2→1)	+287.23; +373.51	−10.02	−4.71	+378.82
β Pic B6†	2013-10-06 06:23	230.528134	CO(2→1)	-9.87; -12.83	+7.63	+20.0	-0.46

†Class A Band 6 window of a different execution block, without a stage-1 flag (Appendix D), shown because it independently shows the same CO component.

sits on this disc's CO belt, peaking near 85 au = 4.3 arcsec at 19.6 pc (Matrà et al. 2017). The window is therefore not a stage-1 outlier because the source fills the ring, which is a physical answer and not a statistical one, and the frozen products already place this emission off the star. A transmitter sits at the star; CO in an edge-on belt sits at the ansae, and the sight line through the stellar position samples the orbit where the radial velocity is near zero. Emission at the star is therefore narrow and centred on systemic while the ansae inside the annulus carry $\pm v_{\rm Kep}$.

Frequencies are compared after removing each block's own tuning, which ALMA sets per date, absorbing most of the barycentric difference; on that basis the Band 3 peaks agree to 1.00 channel and the Band 6 peaks to 3.00 channels. In the barycentric frame, and now in the ordinary radial-velocity convention rather than the frequency-offset sign of Table 6, the CO peaks of the 4 blocks with a recorded epoch span +19.6, +20.2, +20.4, +20.7 km s^{-1}, within 0.7 km s^{-1} of the adopted systemic velocity across two transitions and 8.4 yr, 2013-10-06 to 2022-03-03. Widths place the same conclusion beyond a frequency coincidence: the channel counts above threshold correspond to 10.8–15.2 km s^{-1} in Band 3 and 2.26 km s^{-1} in Band 6, CO linewidths for this belt, where a carrier occupies one channel. The β Pictoris test establishes that the machinery preserves a real repeat; what establishes that *this* repeat would have been seen is the $T_\star = 6.21$ expectation above.

LINE EXCLUSION: DEFINITION, REPAIR, AND COST

The searches as executed used an automated line check matching crossings to catalogued rest frequencies within one spectral channel at zero drift. The β Pictoris windows showed that inadequate: both fell outside its one-channel tolerance, by 10.6 and 21.9 MHz, while lying, after the frame chain, within 3.6 km s^{-1} of the stellar systemic of a resolved, mapped disc, the regime a channel-width tolerance is blind to. Every crossing was therefore reclassified with a velocity-space mask, $\pm 50\,\mathrm{km\,s^{-1}}$ in the stellar frame around each catalogued transition. Its kinematic justification, empirical conservatism and the counter-argument that a communicator might deliberately choose these natural reference frequencies are in Appendix E. The mask is applied *retrospectively*, as a post-hoc classification criterion and never a detection threshold. The half-width was adopted after the β Pictoris crossings had been seen. A sensitivity test reclassifying all 20 crossings at ± 20, ± 30, ± 50 and $\pm 100\,\mathrm{km\,s^{-1}}$ masks 2.1, 3.1, 5.2 and 10 per cent of gross bandwidth and admits no new unattributed flag; at every width the only unmasked unattributed flag is CP−72 2713. Masked regions are reported throughout as *excluded search space*, not as searched-and-empty.

Four defects in the mask as executed are all repaired here. Its rest frequencies were rounded to 1 MHz, worth up to $1.59\,\mathrm{km\,s^{-1}}$. It carried no atomic carbon, though every Band 8 window here is a [C I] $^3P_1$–$^3P_0$ tuning, and no hydrogen recombination line, though H30α falls in the standard Band 6 tuning of much of the sample. It was also applied in the stellar frame alone, the wrong frame for the Galactic foreground behind one of the three line attributions. The repaired catalogue holds 17 transitions of 11 species at full JPL/CDMS precision (Pickett et al. 1998; Müller et al. 2005), including [C I] at 492.160651 GHz and H30α at 231.900928 GHz, against 15 of 9 before, and it is applied in the LSR frame as well as the stellar frame. The species count also reconciles the text with the products, which always used 9 species: earlier versions of this paper said eight, omitting H_2CO.

Re-running the survey was not affordable, so the repaired mask is re-applied to the frozen crossing list as a post-hoc filter, using each crossing channel's own frequency as the pipeline recorded it. **No disposition changes.** The two β Pictoris crossings remain inside the stellar-frame tube, at +0.44 and -0.29 $\mathrm{km\,s^{-1}}$ from CO(1→0) and CO(2→1). HD 48370's crossing remains inside it at $-17.8\,\mathrm{km\,s^{-1}}$ and is now caught by the LSR-frame tube as well, at $-23.9\,\mathrm{km\,s^{-1}}$, which is the frame its foreground attribution always implied. CP−72 2713's crossing lies $-1323.2\,\mathrm{km\,s^{-1}}$ from the nearest masked transition in the topocentric frame and $-1300\,\mathrm{km\,s^{-1}}$ in the LSR frame, so no tube of this mask excludes it; a wider catalogue does place entries inside the tube, which §5.3 treats on the physics. Across all 20 crossing windows the repaired catalogue moves the nearest transition in exactly 1 case, HD 48370's Band 8 crossing at 491.949074 GHz, whose nearest entry changes from CO(4→3) 20098 $\mathrm{km\,s^{-1}}$ away to [C I] at -129 $\mathrm{km\,s^{-1}}$. That is still outside the tube, and the frame corrections are at most a few tens of $\mathrm{km\,s^{-1}}$, so no frame masks it; it was never a flag, its own control ensemble exceeding it. The two new transitions cost a further 0.24 GHz of unique searched frequency, 0.3 per cent.

The masking cost uses one definition: every JPL catalogued transition of the 9 frozen species (276), displaced by each system's radial velocity, $\pm 50\,\mathrm{km\,s^{-1}}$ stellar-frame half-width, intersected with each window and merged. It costs 5.3 per cent of the unique-frequency union (Table 4, Fig. 2) and 3.1 per cent of the window-summed gross bandwidth, the per-window usable fraction having median 0.960 (10th percentile 0.790). CN and CO dominate the loss, and the per-species, per-band breakdown ships in the released veto-mask export. Total exposure is 1.4×10^6 s GHz, or 393 star-hour-GHz, with 30 per cent of it in the five best-populated 2-GHz bins, which any frequency-prior calculation must weight (Appendix D). The mask governs dispositions and leaves the noise-derived thresholds alone, so the quoted $\mathrm{EIRP}_{5\sigma}$ values are unchanged; no signal was found outside the excluded velocity regions in any window.

Masking is a design selection effect: a signal coincident with a molecular line is not thereby "false" but *excluded from the technosignature search domain*, the astrophysical prior there being overwhelming and this pipeline unable to separate the two. We place no constraint on transmitters in the excluded regions, 5.3 per cent of the unique-frequency union. Masked channels are retained for a line-coincident search with independent discrimination. Nor is the exposure confined to the flagged windows: 131 of 431 windows (30 per cent) contain an in-band catalogued transition of the masked species, as do all four flagged windows, yet in none does a crossing fall within one native channel of a rest frequency, the regime the velocity-space criterion covers.

5.4. *Changing the detection statistic, and the AU Mic crossing*

The estimator is justified on its own terms in §4.1, from the exchangeability the rank test needs and independently of any dataset, and the chronology of the change was as follows. An earlier region-max statistic maximised over an $n_{\rm src} = 135$-position region centred on the star against 512 single-position controls, and under it a crossing towards AU Mic in Band 6 at 230.83 GHz was carried as a candidate. The released repository dates both steps. That flag is first committed on 2026 August 31 (`8e00f90e8f8c`), in the manuscript of the day and in its machine-readable aggregate; the asymmetry is identified and the symmetric form of Eq. 2 adopted as operative 9 days later, on 2026 September 9 (`0c465f2661bd`). The redesign postdates the flag, and we do not claim otherwise. The argument therefore rests on the algebra, which owes nothing to these data: under exchangeability an $n_{\rm src}$-position maximum at the star against single-position controls ranks first with probability $n_{\rm src}/(n_{\rm src} + 512) = 20.9$ per cent, and the symmetric form removes that inflation.

Table 7 audits every window the original statistic flagged, so the sequence can be checked. The symmetric criterion flags a strict *subset*, seven before and four after. Both β Pic windows and HD 48370 keep their flags under either statistic; AU Mic, ALMA J1537−3319 and HD 14055 drop out, in each case because the stellar position never exceeded its own control ensemble, the one configuration a localised transmitter cannot produce. Under a decision rule fixed without reference to these data, the 1/513 rank test, AU Mic fails on the same evidence: 10 of its 512 controls reach or exceed the stellar statistic. Because the statistic was settled on these data, the false-alarm calibration here is descriptive of them.

The within-window add-one rank is $p = 0.021$, a different quantity from the survey-wide rate at which any window's ring maximum reaches 5.97 ($p = 0.081$, Appendix F), and far from the 1/513 floor either way. The crossing also

TABLE 7

Every window flagged by the original region-max statistic, under both statistics, so the revision can be audited across all of them, not just the AU Mic window. $S_{\rm reg}$: region maximum of Eq. 2 over $n_{\rm src} = 135$ positions; $T_\star$: the symmetric single-position statistic; 'ctrl max', 'ring max': the largest control statistic under each. The ν column is each window's *centre*; the crossing frequency differs by up to 0.85 GHz, and Table 5 carries $\nu_{\rm cross}$. For CP−72 2713 the region maximum already lay on the star, so the two coincide.

			region-max statistic		symmetric statistic		
Star	Band	ν (GHz)	$S_{\rm reg}$	ctrl max	$T_\star$	ring max	Symmetric outcome
HD 48370	6	230.7305	27.95	26.83	27.10	26.83	flagged
β Pic	3	115.2605	20.31	7.27	14.65	7.27	flagged
β Pic	6	230.5164	12.78	9.12	11.68	9.12	flagged
AU Mic	6	230.5384	5.97	5.85	5.22	5.85	not flagged
ALMA J1537−3319	6	217.0183	5.83	5.63	5.09	5.63	not flagged
CP−72 2713	7	345.1152	5.81	5.68	5.81	5.68	flagged
HD 14055	7	345.1377	5.80	5.74	5.22	5.74	not flagged

fails two pre-named tests, localisation and recurrence. Under the three treatments that decide it, the frozen region-max products, a fully symmetric reprocessing of the same first-epoch block and the second-epoch block of the same programme, the star statistic is 5.97, 4.93 and 4.77 against control maxima of 5.85, 7.70 and 6.78, with add-one star ranks $p = 0.81$ and 0.97 in the latter two.

Appendix F gives both in full: neither the reprocessed first epoch nor the further public Band 6 block at 230.83 GHz shows a localised excess, the second epoch a clean non-recurrence. That rejects a continuous or repeating source at this frequency and these epochs, though not intermittent transmission. Each star was *searched* independently in one to three blocks, so the non-detection concerns the epochs searched and not permanent activity; 62 stars hold further public blocks this survey did not take (§3), and §6 turns that structure into an epoch-activity constraint.

5.5. *Ancillary results*

Two ancillary results are reported in full below; the third ships with the data.

Integration-time audit (machine-readable release). Two of the 58 star–execution-block groups over-count their on-source time against the archive's own exposure accounting, AT Mic by a factor 1.43 and CD−38 10980 by 7 per cent, the rest being conservative. No EIRP threshold depends on integration time, so nothing moves and the exposure-weighted coverage is at worst optimistic by under 1 per cent (Appendix H).

Continuum search (Appendix G). An exploratory by-product over the 57 target/bands processed far enough for a usable measurement: seventeen $\geq 5\sigma$ detections and forty non-detections reaching 0.036–3.6 mJy (median 0.52 mJy). Every detection is astrophysically unsurprising, the auroral emitter LSR J1835+3259 meriting deeper follow-up. No technosignature disposition rests on it.

Serendipitous molecular-line catalogue (machine-readable release; it has no appendix here). The line-exclusion step identifies every $\geq 5\sigma$ outlier channel as a by-product; across the subset scanned, twelve outlier groups are single- or few-channel features with no *masked* transition within 300 MHz and six are plausibly real, all reported openly. That proximity is measured against the 17-transition mask of §5.3, which is what the line-exclusion step compares to, and never against the wider catalogues. §5.3 shows those lie within 60 km s^{-1} of 19 of the 20 crossing frequencies.

6. DISCUSSION

6.1. *What signals could ALMA archival data actually detect?*

Each measured morphology is illustrated with a real window and the completeness entries of Table 8; Fig. 5 maps every window. Denominators count the 81 systems whose deepest threshold in the class lies at or below the case power.

(i) A continuous, frequency-stationary carrier in a coarse channel is the best case. Toward the UV Ceti pair (2.7 pc, deepest window $\mathrm{EIRP}_{5\sigma} = 1.6 \times 10^{13}$ W) such a carrier confined to its channel is recovered at near-ideal $\sqrt{N}$ gain (§4.4), so a carrier matching the *total* EIRP of an Arecibo-like planetary radar (§4.2) is detectable toward the nearest systems at 15.625-MHz resolution. 1 system of 81 qualifies, 2 at twice the power.

(ii) A continuously drifting carrier in a fine 488-kHz channel toward τ Ceti (3.65 pc, threshold 3.8×10^{13} W) is recovered 42 per cent of the time at threshold for $f_{\rm drift} \geq 0.5$ and 75–83 per cent at 1.6–2× threshold (Appendix B). A drifting transmitter above $\sim 8 \times 10^{13}$ W is recovered in 83 per cent of trials with its drift inside the searched ceiling, and 8 systems reach that depth in Class A.

(iii) An intermittent carrier, on for a quarter of the track, toward HD 10647 (17.4 pc, 1.5×10^{14} W) is recovered 91 per cent of the time in fine channels, 100 per cent in coarse ones, so a duty-1/4 transmitter above $\sim 2 \times 10^{14}$ W is detected toward 24 such mid-distance systems. Recovery falls rapidly at smaller dwell fractions (Table 3), and below $f_{\rm dwell} \sim 0.1$ only the nearest windows carry information.

6.2. *What does the null result mean?*

No credible technosignature has been found (§5.3). The $\mathrm{EIRP}_{5\sigma}$ values are nominal trigger powers (§4), and completeness-calibrated exclusion holds only in Class A, where the measured recovery curve applies.

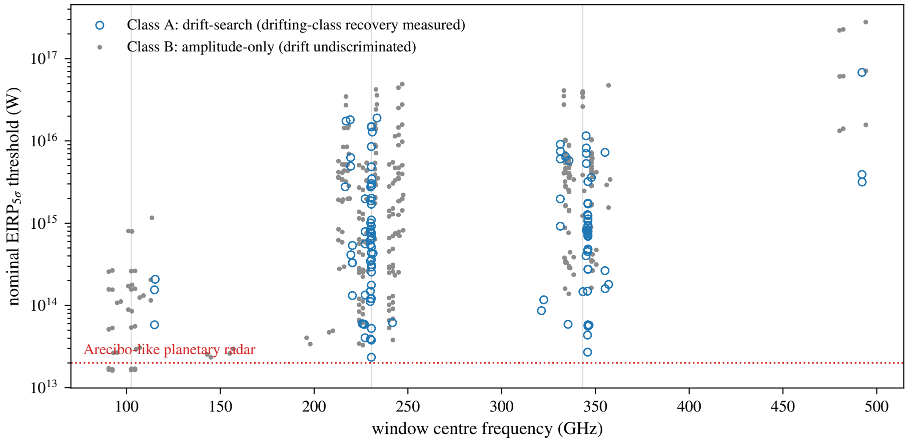

Fig. 5.— Two-dimensional sensitivity: nominal EIRP trigger threshold (boxed rule, §4) against window centre frequency for all 431 searched windows, symbol by channelisation class, the axis along which completeness differs (Table 8). Grey lines mark the Bands 3, 6 and 7 concentrations; dotted, Arecibo-class power.

One exclusion deserves the converse emphasis. The line mask removes $\pm 50\,\mathrm{km\,s^{-1}}$ stellar-frame tubes around the catalogued transitions, whose rest frequencies are the 'magic' channels Cocconi & Morrison (1959) argued a communicator might choose. This survey is blind by construction to a transmitter parked there in the stellar frame, a false-positive control purchased with 3.1 per cent of gross bandwidth and one physically motivated class of signal. Of the 41 planets known around the 18 hosts, only TRAPPIST-1 b reaches the drift ceiling in its edge-on worst case, on star–planet relative accelerations $GM_\star/r^2$ from NASA Exoplanet Archive parameters maximised over orbital phase and inclination. Figure 7 shows the ceiling as a selection function of period and host mass, the host-mass dependence being weak ($a \propto M_\star^{1/3}$ at fixed period). Coverage eligibility and search completeness differ, and an EB-level selection does lie behind the 102 searched blocks: §3 gives the datalink audit that counts 448 public execution blocks in scope against the 102 searched. Those unsearched blocks are the survey's largest single untaken resource, and §6.3 ranks them.

The survey sits against Wright et al. (2018)'s haystack axis by axis. On frequency (10 MHz–115 GHz) only Band 3 falls inside, 20.6 of 115 GHz, 0.18 of the axis, where Hz-resolution surveys go finer; the remaining 72.5 GHz (Bands 4–8, 142–495 GHz) lies wholly outside and is new territory. On transmitted bandwidth (0–20 MHz) a native-channel search probes a fraction set by channel width (mean 0.63). On sensitivity the median threshold of Table 4 detects a 10^{13} W transmitter only within $\sim$0.9 pc. Against the sole prior ALMA-archival search (Mason et al. 2024, 28 bycatch targets at $\geq$1.01 kpc, deepest EIRP$_{\rm min}$ 6.91×10^{17} W), the 81 systems here, 41 inside 20 pc, reach median per-target thresholds $\sim$1250$\times$ deeper. These fractions are not combined into one coverage: the axes are strongly coupled, distance and sensitivity above all, so a product of marginals would misstate the overlap by orders of magnitude.

What the zero detection bounds is conditioned on each target's searched frequencies, the epochs searched and the searched morphology class. Turning it into a fraction of systems requires a per-system detection probability, which is not a demographic occurrence rate. We call it the *conditional searched-domain transmitter fraction* throughout, never "occurrence" or "transmitter fraction" unqualified, and keep its evaluation out of the interpretive argument: the framework, the demonstration and its sensitivity analysis sit in Appendix J, labelled *illustrative calculation only* and among no headline quantity (Table 4). The result of this paper is the search sensitivity, the exposure and the null.

In plain terms: as a by-product of other science, ALMA has stared at 88 of the nearest stars across parts of the millimetre and submillimetre band that no technosignature search had examined. We looked for a narrow, artificial-looking spectral line at each star, and found none the surrounding sky could not also produce. The powers we could have seen are enormous by human standards, and we could look only in the few gigahertz and the few hours each star happened to be observed. A non-detection on those terms says little about how common transmitters are.

Because selection is on archival coverage alone, useful subsamples fall out without compromising the archive-complete framing, subject to the unresolved-pair caveat of §3. 18 of the 88 processed stars (41 known planets) are confirmed exoplanet hosts, among them Proxima Centauri, τ Ceti, ϵ Eridani, β Pictoris and the seven-planet TRAPPIST-1 system; HN Lib and LHS 1140, each with a habitable-zone planet, now carry an ALMA technosignature non-detection. Four searched stars are white dwarfs (Gentile Fusillo et al. 2021): Sirius B, Van Maanen's Star, Wolf 219 and CD−38 10980, all continuum non-detections and among the few mm/submm SETI constraints for white dwarfs (Huang et al. 2026).

6.3. *A next-generation ALMA technosignature search*

ALMA's advantages here are its mm-wave sensitivity, an essentially unsearched tuning range, the interferometric spatial discrimination that supplies the control statistic of §4.1, and a public archive re-processable without new allocations. Six changes would enlarge what this experiment can say, ranked by the signal space each buys.

Search the blocks already held. 346 of the 448 public execution blocks in scope were never downloaded (§3), and one has since settled this survey's only unattributed flag (§5.3). Searching the rest costs no telescope time and would make a multi-epoch survey of 62 stars. The worked case sets the expectation, its two blocks being repeat executions of one scheduling block on one night, so such a pair often supplies a baseline of hours; the distribution of separations is one archive query.

Use the visibilities. The largest unused advantage is the interferometric phase itself: whether a spectral feature carries the signature of a celestial point source at a known position. For each surviving candidate one would compare competing likelihoods, $H_\star$, a source at the propagated stellar position, $V_{ij} = A(t,\nu)\, e^{-2\pi i(u_{ij}l_\star + v_{ij}m_\star)}$; $H_{\rm sky}$, a source elsewhere; $H_{\rm common}$, correlated or common-mode interference; and $H_{\rm noise}$, thermal and calibration residual, and evaluate $\Lambda = \mathcal{L}(V|H_\star)/\mathcal{L}(V|H_{\rm noise}+H_{\rm common})$. That is strictly more powerful than ranking a peak against a control ensemble, and would demote the ensemble to a calibration layer.

Drift discrimination on coarse windows. Within the searched ceiling every in-grid drift on a 15.625-MHz window is sub-channel over a track, so 313 of 431 windows measure amplitude but cannot distinguish a drifting carrier from a stationary one (§4.4). Finer channelisation, or a drift model spanning several blocks, would recover that axis.

Retain more control spectra. The two-stage scheme of §5.3, 512 controls routinely and a local null for any window that passes, fails wherever that null cannot be validated, because full dynamic spectra survive for only 8 control positions. Retaining all of them, at modest storage cost, would make the second stage usable in structured fields too.

Extend the injection campaign and the polarisation axis. The completeness curve should be measured per band, channel width, integration time, antenna count and primary-beam position, and pushed past 10σ until it asymptotes. The design is a stratified grid over those axes, injections at randomised drift rates and channel phases, morphologies extended to sinusoidal acceleration tracks from representative sample orbits, trial-level logging from the first run, and an independent re-implementation cross-check. The polarisation axis buys discrimination against weakly polarised astrophysical backgrounds at no sensitivity cost.

A boundary on drift linearity. The search assumes linear drift over a track. Curvature matters when $\frac{1}{2}\ddot{\nu}T^2$ exceeds half a native channel, reached only by the shortest-period orbits: $P \lesssim 1.6$ d for a typical Class A window (230 GHz, 488 kHz, $T \approx 2000$ s), about an hour for Class B. TRAPPIST-1 b, at 1.51 d, sits just inside the boundary and is also the case exceeding the drift ceiling; elsewhere the constant-drift model is adequate, and the chirp-search extension (Appendix C) covers the remainder.

7. CONCLUSIONS

The public ALMA archive already holds a technosignature search of 88 stars within 40 pc over 93.1 GHz of unique sky frequency between 89.6 and 495.1 GHz, raising a 5σ trigger at powers of roughly 10^{13} to 10^{17} W EIRP. No signal satisfying the survey's statistical, spatial, recurrence and astrophysical-vetting criteria was found. This survey searched, within 40 pc, every star with public usable coverage, 107 target/bands, 88 stars in 81 independent systems, ALMA-archive-complete and audited for position and identity (§5.2), but covering 0.50 per cent of the catalogued stars in that volume and inheriting the archive's non-uniform targeting. The archive's channelisation splits the search. 118 Class A windows cover drift-resolved spectral carriers and 313 Class B windows only unresolved spectral excess, leaving no drift discrimination over 73 per cent of the survey (§4.1). Four windows raised a stage-1 statistical screening flag. Three are consistent with, and very likely caused by, CO emission, two circumstellar components towards β Pictoris across two transitions and three epochs and one previously reported foreground cloud towards HD 48370 (Cataldi et al. 2023). The fourth, towards CP$-$72 2713 at 344.269746 GHz, never acquired an astrophysical attribution and was marginal throughout. A second-stage local null on its own dynamic spectrum gives $p = 1.6 \times 10^{-3}$, about 13 times above the survey-wide Bonferroni scale, and a second public execution block at the same tuning records -1.0 ± 1.8 mJy at the feature's own frequency and drift where the first epoch has 11.3 ± 2.0 mJy. It fails the recurrence criterion pinned before the block was searched and is retired as a candidate, and the data cannot separate a statistical or instrumental excursion from a non-repeating or intermittent event (§5.3). None survives vetting, so we report 0 credible technosignatures. Barnard's Star and Wolf 359 carry their first published mm/submm technosignature limits (§5.1), and forty ancillary continuum measurements (median 5σ limit 0.52 mJy) are non-detections of unquantified power.

These $\mathrm{EIRP}_{5\sigma}$ values are nominal trigger thresholds (boxed rule, §4), 2.3×10^{13}–6.8×10^{16} W over Class A at 15.3–1953 kHz native resolution and 1.6×10^{13}–2.8×10^{17} W over Class B at 15.625–31.25 MHz, deep enough for an unresolved carrier matching the total power of an Arecibo-like planetary radar toward 1 of the 81 systems. The conditional searched-domain transmitter fraction stays illustrative (Appendix J). The *Scope of the constraints* box of §4 fixes what the non-detection binds, and outside it nothing is bounded. §6.3 ranks the extensions that would enlarge it. By-products ship with the data, a serendipitous molecular-line catalogue, exoplanet-host and white-dwarf subsample limits, and a closure-phase diagnostic demonstrated without discriminating power at these flux densities.

ACKNOWLEDGEMENTS

This paper uses ALMA data from the public ALMA Science Archive. The 102 execution blocks searched, with project codes and epochs, are listed in the machine-readable release and should be cited from it in the standard JAO.ALMA#XXXX.X.NNNNN.S form. They span 2013 October 6 to 2025 June 11 across 36 proposal codes. The AU Mic crossing window of §5.4 comes from the 2014 August 18 block of project 2012.1.00198.S, and one further public block of that project (uid://A002/X7d80c7/X15c6, 2014 June 5) also covers the 230.83 GHz window and is the second epoch of its recurrence test. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), MOST and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (CDS, Strasbourg, France).

COMPETING INTERESTS

G.J.W. declares no competing interests. R.D. is an employee of VBRL Holdings Inc. (a private software company with no astronomy or SETI-related product line), which provided no funding for and took no role in this work. The authors declare no competing interests.

AUTHOR CONTRIBUTIONS (CREDIT)

Glenn J. White: Conceptualisation, Methodology, Supervision, Writing – review & editing.
Robin Dey: Software, Investigation, Formal analysis, Data curation, Visualisation, Writing – original draft.

DATA AVAILABILITY

The reduced data products underlying this paper, with the full target-selection and analysis pipeline, are at `https://github.com/Tilanthi/SETI`. All numerical values here come from the *submitted-analysis snapshot*, the repository commit tagged at submission, which freezes pipeline code and input data as run and reproduces every number, table and figure from the regeneration scripts shipped with it.

The machine-readable catalogue, `per_target_results_v3.47.csv`, carries 431 rows, one per searched window, in 41 columns. Each row gives star name and machine-readable `system_id` (the independent stellar system that is the statistical unit, seven bound pairs sharing one identifier), distance, ALMA band and execution block, searched frequency range, native channel width, on-source time and integration count, per-channel rms, the nominal $\mathrm{EIRP}_{5\sigma}$ threshold with both Hanning-corrected values (§4), the drift ceiling as a rate and as an acceleration, the trial drift count, η_{drift}, η_{smear}, resolution and search class, stellar and control peak statistics with the count of controls at or above the star and the add-one rank probability, the crossing frequency, the nearest catalogued transition with its offset in MHz and $\mathrm{km\,s^{-1}}$, the full control geometry (ring centre, θ_{PB}, r_{in}, r_{out}, control count, seed) and the disposition, verbatim from Table 5 and so carrying the CP−72 2713 retirement. The re-searched β Pictoris blocks of §5.3 lie outside this frozen release; their per-block $T_\star$, control maximum, control count, channel width and epoch ship in the accompanying epoch-extension record. Four defects disclosed in earlier versions are repaired: HD 139084B is no longer counted as two systems, so 81 is the independent-system total; the empty `n_ant` column is removed, per-window antenna counts being recoverable only from the raw measurement sets; the disposition strings match Table 5; and the directory-name sanitisation that stripped the sign from four designations is undone, so `star_name` now reads LSR J1835+3259, PM J03433+1958, WD 0407−179 and BD+05 1668.

The complete release will also be deposited on Zenodo, with a DOI minted on final acceptance and inserted here. It comprises (i) per-window search products for all 431 windows: spectral extracts, per-drift-trial peak statistics, the machine-readable line-exclusion mask in both its frozen and repaired forms (§5.3), archive UID, epoch-propagated target coordinates and field centre, primary-beam gain, channel frequencies and widths, the drift grid, and the target statistic with all 512 control statistics from which the rank and p-value are recomputable; (ii) control-ensemble calibration products at all 512 positions; (iii) closure-phase and population-anomaly outputs; (iv) the full injection-recovery campaign products, all 1200 trials of Appendix B and all 3 888 dwell trials, together with the second-stage local-null code, its per-window outputs and the retained null-draw arrays (§5.3); (v) the frozen pipeline code and input data snapshot with its change-log; (vi) the execution-block metadata and integration-time audit tables; (vii) the serendipitous molecular-line catalogue, `line_catalogue.json`, written by the pipeline's own continuum line-exclusion step; (viii) the execution-block availability census of §3, from the ALMA datalink service, with its per-star ledger and the annulus-geometry and smearing-census table fragments generated for this paper; and (ix) the retained per-integration dynamic spectra behind the time-domain and continuum checks of §5.3, which are held on the processing host rather than in the manuscript source bundle; and (x) the second-epoch search products behind the CP−72 2713 retirement (§5.3), the four `A002_Xff0235_X502d_spw{0..3}` result, search and source-spectrum files, together with `cp72_recurrence_v345.json` and the script that derives it. That block was declined during the survey by the per-target download budget of §3, at 49.5 GB against 49.3 GB for the block that was taken. Data products carry the Creative Commons Attribution 4.0 International licence (CC BY 4.0), and the pipeline code the MIT licence; neither author's employer asserts any intellectual-property claim over the released code. Raw ALMA visibility data are public from the ALMA Science Archive at `https://almascience.org`.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; calibration, imaging and the local reprocessing of §§5.3–5.4), Python 3 with astropy, numpy and scipy (spectral extraction and statistics), and the ALMA Science Archive and SIMBAD TAP interfaces; all are free and open-source, with exact versions recorded in the frozen release.

APPENDIX

A. FREQUENCY CONVENTIONS, LINEWIDTH READING, AND SMEARING

Reference frames. The line mask's chain runs topocentric → barycentric (Earth ephemeris at the execution-block mid-time) → stellar rest frame (catalogued systemic radial velocity), with JPL/CDMS rest frequencies (Pickett et al. 1998; Müller et al. 2005); in the LSR frame it stops at the standard solar motion, not the systemic velocity

(§5.3). The residual error is the catalogue radial-velocity uncertainty alone, a few $\mathrm{km\,s^{-1}}$ within 40 pc; second-order relativistic terms at the $50\,\mathrm{km\,s^{-1}}$ scale are $\lesssim 5$ kHz at 350 GHz, below the finest native channel searched (15.3 kHz). The β Pictoris and AU Mic attributions use the same chain, so their offsets are comparable to the exclusion tolerance.

Channel widths, and the instrumental response. These come from the measurement sets' own spectral window tables (CHAN_WIDTH). Earlier versions read a missing "EFFECTIVE_BANDWIDTH" column as evidence of no online spectral averaging; MS v2 defines no such column, so the check is withdrawn, and with it the claim that the quoted channelisations are noise bandwidths.

ALMA applies online Hanning smoothing by default, so the spectral response is $\sim$2 channels wide and neighbouring channels are correlated. For the 0.25/0.5/0.25 kernel the coefficients are $\rho_1 = 2/3 = 0.667$ and $\rho_2 = 0.167$, with $\sigma_{\rm smoothed}/\sigma_{\rm raw} = 0.612$: the correlation is not confined to adjacent channels, the lag-two term being a sixth. Under the 0.25/0.5/0.25 kernel a sub-channel tone keeps 0.50 of its power in the peak channel at the channel centre and 0.38 on a boundary, median 0.44 over uniform placement. Every $\mathrm{EIRP}_{5\sigma}$ here is therefore optimistic by $\times$2.00 to $\times$2.7, median $\times$2.29; the released catalogue carries the nominal threshold and both corrected values per window, and the sensitivity summary quotes the worst case. The injections deposit an unsmoothed delta-in-channel tone, so the recovery curves of §4.4 and Appendix B carry the same bias and the end-to-end check (Eq. 4) cannot see it. The retained per-integration residuals of the second-stage local null (§5) have a measured lag-one spectral autocorrelation of 0.65, against the Hanning prediction 0.667, so that correlation is instrumental and the residual spectra are smoothed and un-averaged, which is the independent confirmation of the smoothing state. The archive's reported effective spectral resolution per window, against each measurement set's own channel separation, is exactly 2.00 in 403 of the 431 windows, in both correlator modes and including 309 of the 313 coarse time-division-mode (TDM) ones; the other 28 report a smaller ratio, the signature of online channel averaging, where the peak-channel loss is smaller. The blanket factor is right for the great majority of both classes. We still do not rescale, so each threshold here is nominal and up to $\times$2.7 optimistic.

Linewidth convention. The statistic is the single-channel flux density after drift correction, so the quoted $\mathrm{EIRP}_{5\sigma}$ applies to any transmitter narrower than the native channel, flat in the assumed linewidth, with two sub-channel qualifications. Flatness is exact only for a centred tone, a boundary-straddling one splitting its power between two channels at worst by a factor of two; and the channel-to-channel transition follows the correlator's passband shape, which the frozen products do not retain, so flatness is a channel-centred idealisation good to tens of per cent. Thresholds therefore carry a sub-channel-position systematic of that size, a randomly placed tone keeping a mean 0.75 of its power in the stronger channel, and EIRP values are quoted to two significant figures.

Transmitters wider than one channel. The baseline filter (§4.1) is flat for features narrower than roughly half its effective width W and falls to zero above W, so the survey's sensitivity is to channel-confined morphologies and W is the linewidth ceiling: 65–74 channels over the fine class and 30–59 over the coarse one, that is $\approx$32 MHz at 488 kHz and $\approx$0.9 GHz at 15.62 MHz. Within that ceiling, a top-hat transmitter spanning $N_{\rm ch}$ native channels has per-channel flux the total divided by $N_{\rm ch}$ at unchanged noise, so to first order

$$\mathrm{EIRP}_{5\sigma}(N_{\rm ch}) \;\simeq\; N_{\rm ch} \times \mathrm{EIRP}_{5\sigma}(1), \qquad N_{\rm ch} \lesssim W/2, \tag{A1}$$

the rescaling the main text applies to signals broader than one channel.

Worked example: from flux to EIRP. Take UV Ceti (G 272-61B) Band 3, the deepest of its 8 windows in that band (execution block A002_X877e41_X14c1, lower edge 91.565 GHz). Its per-channel rms is 0.241 mJy, so $S_{\rm min} = 5 \times 0.241 = 1.21$ mJy in one 15.625-MHz native channel, or $1.21 \times 10^{-29}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$ at 2.67 pc, and $\mathrm{EIRP}_{5\sigma} = 4\pi d^2 S_{\rm min}\Delta\nu = 1.6 \times 10^{13}$ W.

Intra-integration smearing. The frequency excursion at the maximum searched drift rate across one integration, $\dot\nu_{\rm max} t_{\rm int}$, is at most 1.10 channels across the 431 windows (median 0.001, 90th percentile 0.06). A linear drift smeared by Δ channels within an integration keeps a fraction $\eta_{\rm smear} = \mathrm{sinc}(\pi\Delta/2)$ of the peak-channel amplitude, so the effective threshold is $\mathrm{EIRP}_{5\sigma}/\eta_{\rm smear}$. The 9 windows with $\eta_{\rm smear} < 0.99$, generated from the released catalogue as star, band, ($\eta_{\rm smear}$, threshold multiplier): HIP 10679 Band 6 (0.574, $\times$1.74), β Pic Band 6 (0.574, $\times$1.74), HD 172555 Band 6 (0.574, $\times$1.74), HIP 10679 Band 6 (0.606, $\times$1.65), HD 172555 Band 6 (0.606, $\times$1.65), η Crv Band 8 (0.865, $\times$1.16), η Crv Band 7 (0.932, $\times$1.07), HD 48370 Band 6 (0.979, $\times$1.02) and CP$-$72 2713 Band 6 (0.979, $\times$1.02). The remaining 422 agree with their nominal thresholds to better than 1 per cent (compounding the worst smearing penalty with the worst Hanning factor gives $\times$4.6 on the nominal threshold for those five windows; no released column carries the product). The $\mathrm{sinc}(\pi\Delta/2)$ form is the transform of a tone swept uniformly across Δ channels within one integration, evaluated at the peak channel, and is the conservative reading: a pure occupancy argument would retain $\min(1, 1/\Delta)$, giving 1.0 rather than 0.64 at $\Delta = 1$. A signal drifting midway between adjacent trial rates accumulates at most 0.03 channels of residual drift across the grid as realised, so no significant sensitivity is lost between them.

B. INJECTION-RECOVERY SENSITIVITY VALIDATION

To test whether a 5σ threshold behaves as idealised Gaussian noise predicts, we inject synthetic tones of known amplitude into real visibilities, through the extraction code's own mechanism and as far upstream of the statistic as retained products allow, and measure what the unmodified pipeline returns. Three results inherit its single-configuration limitation: the drifting-class recovery of Appendix J, the machinery and boundary-placement amplitude factors of §4.3, and the coarse-window dwell reading of §4.4.

Six configurations span three bands and both classes: three coarse Band 7 spws of AT Mic, one coarse spw each in Bands 3 and 6 of AU Mic, and AU Mic's 488-kHz flagband window with its 49-trial drift grid. Each receives a complex

TABLE 8

Amplitude completeness at the trigger, by window class and carrier morphology, from the 1200 injections of this appendix. "Detection" is amplitude recovery at the nominal trigger; "drift" is whether a recovered drift rate is informative about the transmitter; N counts the searched windows of that class. The second experiment, Table 3, is *not* a threshold completeness: it injects non-drifting carriers at $\geq 4\sigma$ per integration into 12 real windows and measures temporal occupancy. The two must not be read across.

Class	Morphology	Detection	Drift	N
Class A (fine)	drifting	measured: $C_{5\sigma} = 0.42$	measured	118
Class A (fine)	near-static	post-baseline only[a]	n/a[b]	118
Class B (coarse)	any	end-to-end	none	313

[a] The dwell campaign recovers 91–100 per cent at $f_{\rm dwell} \geq 0.25$, injecting after the baseline step; through-pipeline recovery is unmeasured.
[b] Sub-channel drift over the track by definition.

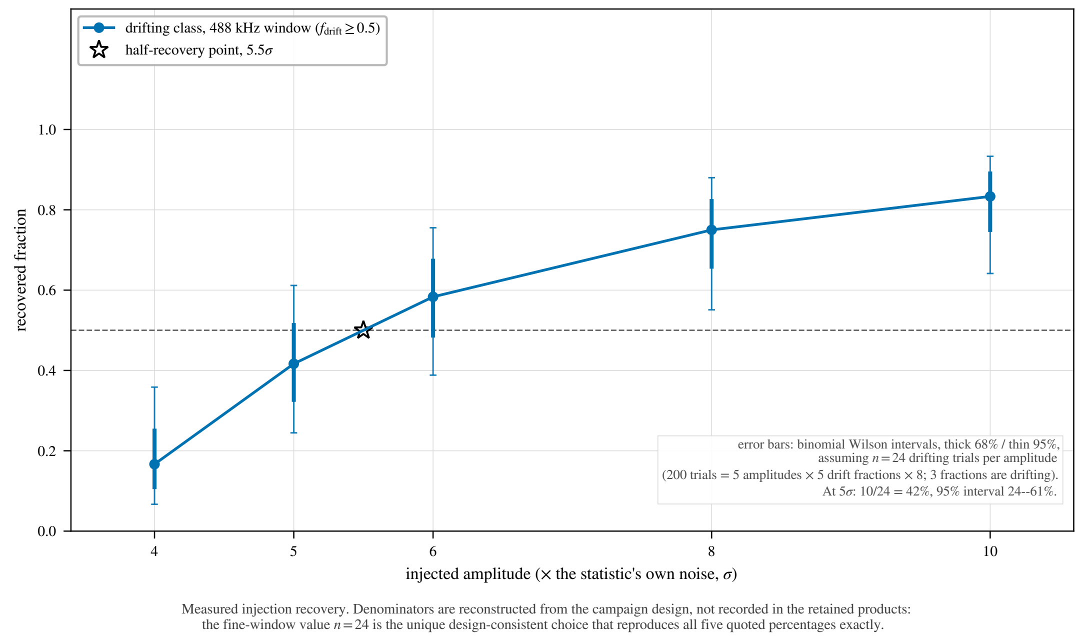

Fig. 6.— Measured injection recovery on real ALMA data, fine-channel configuration. Recovery $C(A, f_{\rm drift})$ rises with amplitude for drifting carriers; static and near-static ones fail the frozen criterion, and boundary-straddling tones can fail its peak-drift clause while still being detected. Coarse configurations are not plotted: under the same criterion they recover nothing, withdrawn in the text as an artefact of the criterion.

tone at the star position, at amplitudes 4–10× the statistic's own noise, drift fractions $f_{\rm drift} = 0$, 0.25, 0.5, 0.75, 1.0 of the ceiling, on- and half-grid rates, and channel-centred and boundary placements: 200 trials per configuration, 1200 in all, under a frozen criterion ($T \geq 5$, peak channel within 3 of the injection, peak drift within one grid step).

The coarse-window variant recovered 0 of 500 injections at every amplitude from 4σ to 10σ under that criterion. The recovery criterion produced that artefact and the pipeline suppresses nothing, which makes it the one withdrawn result here: the drift-matching clause cannot be satisfied on coarse windows, where every in-grid drift is sub-channel over the track, so the reported peak drift is arbitrary. Scored on detection alone the same windows recover essentially everything, and an end-to-end injection through the released pipeline returns $T_\star$ of 20.5, 65.3 and 217.7 at 1, 3 and 10 times the per-integration noise, against an ideal coherent gain of $\sqrt{639} = 25$. Class B limits therefore bound carriers of any dwell fraction, with the completeness of §4.4; what those windows lack is drift discrimination. The fine window fails the same criterion the same way, and that cell is unmeasured, since carriers with $f_{\rm drift} \leq 0.25$ dwell $\geq$40 per cent of the track per channel and fail it, while those with $f_{\rm drift} \geq 0.5$ survive as the searched class.

The second result is the completeness curve. On the fine window, drifting carriers ($f_{\rm drift} \geq 0.5$, pooled over grid phase and sub-channel placement) are recovered 17, 42, 58, 75 and 83 per cent at 4, 5, 6, 8 and 10σ, saturating by twice threshold power (Fig. 6). The machinery alone, with the tone injected after the baseline step, is amplitude-faithful to within $-9/+14$ per cent for channel-centred tones at every drift fraction. Boundary-straddling placement costs a factor 0.6–0.8 in the statistic and recovers 0 per cent at 5σ against 83 per cent for centred ones (67 and 100 per cent at 10σ), crossing 50 per cent near 1.4× threshold amplitude. Survey completeness uses the placement-pooled curve, conservative for the boundary-straddling case: $\mathrm{EIRP}_{50} = 1.10\,\mathrm{EIRP}_{5\sigma}$ and $\mathrm{EIRP}_{90} > 2.0\,\mathrm{EIRP}_{5\sigma}$, a bound only, both carrying an unbounded transfer systematic to every configuration but this one, bracketed as 0.5–2× in Appendix J. Not spanned: other target environments, primary-beam offsets, integration durations and edge-phased drift grids; bounding that transfer error, propagated in Appendix J, needs the multi-configuration campaign of §6.3.

TABLE 9

SELECTION FUNCTION OF THE SEARCHED SAMPLE AGAINST A VOLUME-LIMITED REFERENCE CENSUS. REFERENCE CENSUS: GAIA DR3 SOURCES WITH $\varpi > 0$, $\sigma_\varpi/\varpi \leq 0.1$ AND $1/\varpi \leq 40$ PC (17566 OBJECTS; 8594, 49 PER CENT, CARRY A `teff_gspphot` ESTIMATE). SAMPLE: THE 88 STARS WITH AT LEAST ONE SEARCHED WINDOW, AT THE DISTANCES THE SEARCH ITSELF USED. TEMPERATURES ARE AVAILABLE FOR 52 OF THE 88 STARS (ALL 88 MATCHED TO THE 168-ENTRY ALMA-COVERED CENSUS, 58 BY EXACT NAME; 52 OF THOSE MATCHES CARRY A TEMPERATURE), SO THE TEMPERATURE PANEL IS NORMALISED WITHIN EACH COLUMN OVER CLASSIFIED OBJECTS ONLY, WITH $T_{\rm eff}$ A PROXY FOR SPECTRAL CLASS. "RATIO" IS THE SAMPLE COLUMN FRACTION OVER THE CENSUS COLUMN FRACTION: 1 MEANS THE SAMPLE TRACKS THE REFERENCE POPULATION, >1 OVER-REPRESENTATION.

Property	Census	Searched	Ratio
Distance (all 88 searched stars; 17566 census objects)			
$d \leq 5$ pc	48 (0.3%)	12 (13.6%)	49.9
5–10 pc	240 (1.4%)	13 (14.8%)	10.8
10–20 pc	2069 (11.8%)	18 (20.5%)	1.7
20–30 pc	5391 (30.7%)	18 (20.5%)	0.7
30–40 pc	9818 (55.9%)	27 (30.7%)	0.5
$T_{\rm eff}$ class[a] (52 searched; 8594 census)			
A (≥ 7500 K)	24 (0.3%)	1 (1.9%)	6.9
F (6000–7500 K)	402 (4.7%)	9 (17.3%)	3.7
G (5300–6000 K)	860 (10.0%)	16 (30.8%)	3.1
K (3900–5300 K)	1400 (16.3%)	5 (9.6%)	0.6
M (< 3900 K)	5908 (68.7%)	21 (40.4%)	0.6
Other axes			
Known planet hosts[b]	20 (11.9%)	18 (20.5%)	1.7
Searched stars / systems	–	88 / 81	–

[a]Bins are temperature bins, not MK types: A ≥ 7500, F 6000–7500, G 5300–6000, K 3900–5300, M < 3900 K. Gaia `teff_gspphot` is unavailable for about half the reference census, preferentially for the coolest and faintest objects, so the census M fraction is a lower limit and the earlier-type ratios are correspondingly upper limits. White dwarfs are not separable in this parameterisation and fall wherever their (unreliable) photometric temperature places them.

[b]The Gaia reference census carries no planet information; the census column here is instead the 168-entry ALMA-covered census, of which 20 entries are known planet hosts. The 88 searched stars include 18 hosts of 41 known planets.

C. ANCILLARY SCREENS WITH NO OPERATIVE ROLE HERE

Closure phase. The closure phase $\arg(V_{ij}) + \arg(V_{jk}) - \arg(V_{ik})$ vanishes for an unresolved point source anywhere on sky and is immune to per-station calibration errors, so it tests point-source coherence alone and a coherent far-sidelobe interferer would pass. We score consistency with zero by a circular z-score of the mean drift-tracked closure phase, nulled on each observation's own integrations, and ran it on continuum detections in four local reprocessing fields, the resolved AT Mic Band 7 binary among them as a control. At these flux densities it is useless: per-baseline S/N is 0.5–0.7 at the median, only 0.02 per cent of triangles have all three baselines above 5, and every field's closure phases, the binary included, are indistinguishable from noise.

Chirped drift and periodicity. Stacking at constant linear drift rate is blind by design to a drift rate that changes during the observation, and to pulsed or periodically modulated carriers. We implemented a chirped (quadratic-drift) extension on a coarser grid of 21 linear-drift $\times$ 5 chirp-rate trials, and a per-channel periodicity search, a per-position FFT along each channel's time axis; both vet the star against 8 retained control positions, a strict subset of that window's ensemble. On the 122 retained spectra this is a demonstration: flag rates of 10.7 and 13.1 per cent against a $\sim$11 per cent chance rate show only that the extension runs correctly under the null, and no limit derives from it.

Cross-target frequency occupancy. With no ON/OFF cadence against radio-frequency interference, this single-pointing archival survey substitutes the many unrelated stars sharing similar correlator setups: a channel-confined feature recurring at the same observed *sky frequency* towards different targets is almost certainly interference or instrumental. Over every target/spw result the check pairs the best-fit-channel frequency and the formal $\geq 5\sigma$ crossings, matching ν_1, ν_2 towards different targets when $|\nu_1 - \nu_2| < \max(k\,\Delta\nu_1,\ k\,\Delta\nu_2)$ with $k = 3$ ($\sim$46 kHz). Across the 431 windows and 107 tabulated crossings, no pair falls within the kernel, or within 0.5 MHz. The kernel is narrower than the β Pictoris line offsets of 10.6 and 21.9 MHz, so those windows were dispositioned by line-catalogue inspection.

The null's power is the count of target pairs sharing frequency coverage: on the $\leq$20 pc subset, 366 of 946 star pairs overlap by at least 0.01 GHz and 109 involve one of the five stars carrying formal crossings, so every crossing frequency was tested against overlapping independent targets. The UV Ceti and AT Mic shared execution blocks are excluded as identities by construction. The interim product's per-window crossing lists are capped at 50 frequencies, so the true count is 171 crossings against 107 tabulated. The difference is one window, β Pic Band 6 ultra-fine, whose 114 formal crossings enter as the 50 strongest; its untabulated 64 could not be tested, so the null is demonstrated for the tabulated set and unresolved for the rest. The public release ships uncapped crossing lists.

D. PER-TARGET RESULTS

The per-window results ship as the comma-separated catalogue of the Data Availability statement. Its $\mathrm{EIRP}_{5\sigma}$ column uses the definition of §4, on the same threshold and primary-beam correction as the continuum column, so a row's two limits are comparable; its per-channel flux-density threshold uses the carrier lane's per-channel noise rather than the tabulated `rms_mJy`, the two differing by $5 \times G(r)$. Bands 6 and 7 supply 86 per cent of the windows.

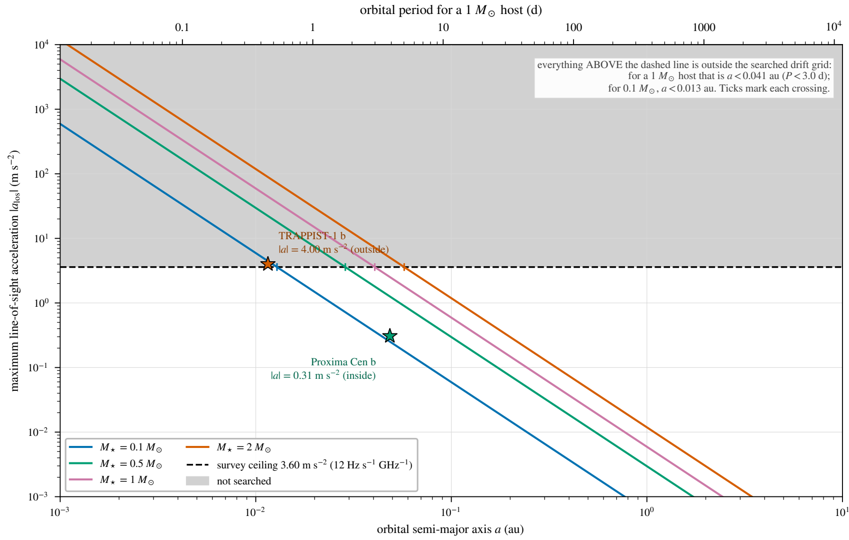

FIG. 7.— The drift-rate ceiling as a physical selection function. Maximum star–planet relative acceleration, edge-on and maximised over orbital phase, as the carrier drift rate it induces (Hz s^{-1} per GHz of carrier frequency; equivalently $\dot{\nu}/\nu = \dot{v}/c$), against orbital period, for host-star masses 0.09–1.8 $M_\odot$ (lines) and the known planets of the planet-hosting targets (points). Shaded band: the survey's drift-search ceiling, above which a transmitter lies outside the searched grid. Only TRAPPIST-1 b reaches it (§6).

TABLE 10

PER-BAND SURVEY CHARACTERISATION OF THE 431 SEARCHED WINDOWS. "SYS." COUNTS UNIQUE STELLAR SYSTEMS SEARCHED IN THE BAND, SO A SYSTEM OBSERVED IN TWO BANDS IS COUNTED IN BOTH ROWS (81 SYSTEMS, 88 STARS IN TOTAL); "EBS" THE CONTRIBUTING EXECUTION BLOCKS, ATTRIBUTED BY THE PROVENANCE OF THE DE-DUPLICATED WINDOWS; $\Delta\nu_{\rm ch}$ THE NATIVE CHANNEL WIDTH IN MHZ (MEDIAN, RANGE IN BRACKETS); "UNION" THE UNIQUE SKY FREQUENCY SEARCHED IN THE BAND AFTER MERGING OVERLAPPING WINDOWS; EIRP$_{5\sigma}$ THE MEDIAN NOMINAL THRESHOLD ACROSS THE BAND'S WINDOWS; $\dot{\nu}_{\rm ceil}$ THE RANGE OF DRIFT-RATE CEILINGS IN HZ S^{-1} PER GHZ OF CARRIER FREQUENCY. THE "ALL" ROW IS COMPUTED OVER THE WHOLE SAMPLE, NOT SUMMED DOWN THE COLUMNS: EBS AND SYSTEMS RECUR ACROSS BANDS AND THE FREQUENCY UNIONS ARE DISJOINT BY CONSTRUCTION.

Band	Sys.	EBs	Win.	$\Delta\nu_{\rm ch}$ (MHz)	Union (GHz)	Med. EIRP$_{5\sigma}$ (W)	$\dot{\nu}_{\rm ceil}$
3	8	9	40	15.63 (0.244–15.63)	20.6	1.1×10^{14}	12.0–13.3
4	1	1	4	15.63	6.9	2.6×10^{13}	12.0
5	1	1	4	15.63	6.9	4.4×10^{13}	12.0
6	54	54	216	15.63 (0.015–31.25)	29.4	1.3×10^{15}	12.0–13.3
7	33	34	155	15.63 (0.061–15.63)	22.6	2.9×10^{15}	12.0–13.3
8	3	3	12	15.63 (0.061–15.63)	6.6	6.1×10^{16}	12.0
All	81	102	431	15.63 (0.015–31.25)	93.1	1.4×10^{15}	12.0–13.3

E. THE VELOCITY-SPACE LINE MASK: JUSTIFICATION AND RETROSPECTIVE APPLICATION

The one-channel line check failed the β Pictoris windows (§5.3): a native channel spans a fraction of a km s^{-1}, while circumstellar kinematics shift real emission by many channels. The replacement mask takes its tolerance from the kinematic budget of circumstellar gas: Keplerian motion where these molecules survive ($\gtrsim$10 au), $\lesssim$13 km s^{-1}; cold-gas linewidths $\lesssim$1 km s^{-1}; cold dense-gas tracers with no bulk-wind term for the dwarfs dominating the sample; and an uncorrected barycentric term bounded by $\pm$30 km s^{-1}. The result is empirically conservative: the largest attributed circumstellar component (β Pic, within 0.46 km s^{-1} of systemic) sits at 0.9 per cent of it, and *zero* crossing frequencies lie 13–50 km s^{-1} from their nearest *masked* transition (the wider catalogues are another matter: §5.3), so a $\pm$13 km s^{-1} Keplerian-only mask would have suppressed and released the same windows: the wider tolerance costs no yield and buys robustness against radial-velocity catalogue error.

The repaired mask. The four defects and the retrospective re-application that repairs them are in §5.3; no disposition changes. The rounding defect is why the released CO(1→0) offset column, referred to the frozen mask's rounded 115.271000 GHz entry, reads 0.202 MHz (0.53 km s^{-1}) smaller than the laboratory-referred value of Table 6. A cheaper, better-founded rule than enumerating species: the archive records the rest frequency each execution block was tuned to, so masking each window's own tuned line would catch the β Pictoris, HD 48370 and [C I] windows automatically, alongside an LSR-frame tube of a few $\times$10 km s^{-1}.

The crossing ledger, once: of the 20 windows with an on-star crossing, 4 also beat their control ring and three of those are accounted for by the mask. Two further crossing windows fall inside the mask without beating their ring (the unflagged β Pic Class A window at -0.46 km s^{-1}; HD 285968 at +8.7 km s^{-1}, already rejected by its ring, maximum 10.9 against a star peak of 6.5), and the remainder lie far outside any circumstellar tolerance (AU Mic +379, LHS 1140 +115 km s^{-1}). The other 16 are ordinary noise crossings: in each, the window's own control ring reaches or exceeds the stellar value (ring-maximum median 5.84 against a crossing median 5.18), the expected Gaussian-tailed population of moderate 5–5.5σ crossings with comparably strong peaks among 512 controls. But the transitions that make astrophysical false positives cheap are also natural reference frequencies for a deliberate communicator (Cocconi

& Morrison 1959), so the released products keep a known-line-coincident list for a future search with independent discrimination: every on-source 5σ crossing inside the masked intervals, with its offset from each nearby transition.

F. AU MIC: THE THREE TESTS IN FULL

Localisation. Reprocessed from the raw science data model with the observatory pipeline's own calibration recipe (310 integrations against the release pipeline's 249, on a finer drift grid) and re-run through the symmetric statistic of §4.1, the crossing window's star does not beat its control ensemble. The frozen products agree only asymmetrically: the source-region peak exceeds the 512-control maximum by 0.12, below that statistic's window-to-window scatter, and the crossing fails the single-position variant too.

Recurrence. The one further public Band 6 block covering 230.83 GHz, the same programme executed 2014 June 5, was searched identically: no channel within ± 10 of the first-epoch flagged channel exceeds 3.1, and nothing is localised to the star. Its band rms (1.48 mJy per 488-kHz channel) matches the first epoch's 1.9 mJy, so a comparable excess would have been recovered, and the frequency recurs at no other target. Intermittent or precisely-scheduled transmissions outside the sampled windows stay unconstrained.

Statistical scale. The on-source peak (5.97) sits at the 92.1th percentile of the control-maxima distribution over all 431 windows (median 4.56, 90th percentile 5.85): 34 windows (7.9 per cent) match or exceed it, add-one empirical $p = 0.081$ before any multiple-testing correction: a survey-wide window-level rate, distinct from the within-window rank of §5.4. The crossing matches no catalogued molecular transition (Table 6), and closure-phase vetting finds it point-source consistent, which is one-sided evidence (Appendix C).

Two astrophysical contexts are recorded for re-examination; neither grounds the rejection. AU Mic is an exceptionally active flare star, but the block's own quiescent continuum (0.15 mJy) rules out the later-epoch GHz-wide flares of MacGregor et al. (2020) (16.8 and 6.1 mJy): an 11 mJy channel-confined excess would sit $\sim 70\times$ above a continuum that shows nothing. The controls share the primary-beam footprint, so smooth structure such as the debris belt raises star and controls alike and cannot alone produce a localised excess.

G. THE ANCILLARY CONTINUUM LANE

The continuum lane is an exploratory by-product, with 57 measurements, 17 detections and 40 limits, a millimetre excess being at best an indirect, model-dependent observable of waste-heat technology. An "upper limit" is $5\times$ the primary-beam-corrected map rms at the star's position, and the deepest non-detection is towards TRAPPIST-1 in Band 3. Under the combined error model sixteen of seventeen detections stay above 5σ; the exception (η Corvi Band 7, 4.7σ combined) is marginal. Bright far-from-target peaks fall outside the 3-arcsec tolerance and count as non-detections.

Every detection is astrophysically unsurprising: the unresolved UV Ceti binary; six systems of disc or chromospheric emission (β Pic, AU Mic, AT Mic and τ Cet among them); six stars at photospheric flux; and two late-M dwarfs of documented magnetic activity, LP 349−25 and the auroral emitter LSR J1835+3259 (Hallinan et al. 2015). Closure phase discriminates structure for none of them at these flux densities.

The anomaly screen. Each star's flux is tested against its own photospheric expectation (Planck function with Gaia DR3 parameters, Pecaut & Mamajek fallback where GSP-Phot is unavailable) and against same-$T_{\rm eff}$ stars in coarse bins of at least five, by a robust median/MAD statistic. The error model, evaluated before any flag, is $\sigma^2_{\rm tot} = \sigma^2_{\rm map} + (f_{\rm cal}\, S_\nu)^2 + \sigma^2_{\rm model}$, with $f_{\rm cal}$ 5 per cent in Bands 3–6 and 10 per cent in Bands 7–8 and $\sigma_{\rm model}$ propagated from Gaia temperature and radius errors and the fallback relation's scatter. The model has *no intrinsic-variability term*, which is adequate for a photosphere and wrong for an active M dwarf whose millimetre flux can vary by factors of several, so the screen is uninformative for exactly the stars a radio-stellar reader cares about, and at the five-star minimum it flags for inspection, without testing per-star hypotheses. One outlier results: χ^1 Ori at Band 3 (excess ratio 2.52, resampled tail probability 0.53 in its six-star bin), consistent with a spectroscopic G0V binary's unresolved companion or a chromosphere.

H. FALSE-ALARM ACCOUNTING UNDER THE SYMMETRIC CRITERION

The per-window rate is 1/513, the survey expectation $431/513 = 0.84$ flags. The floor falls only if the control ensemble grows, which the primary beam bounds; and windows are not fully independent, grouping into 102 execution blocks that share weather, calibration and correlator setup, so $N/513$ is an expectation over a mildly correlated sample, not N independent trials. Both limits are conservative for the conclusion drawn, that no window warrants candidate status: they widen the null range.

The trials budget, in one set of units. A window holds 177–6 866 208 channel×drift cells (median 468), and the survey 4.60×10^7 in total. A one-sided Gaussian 5σ tail (2.87×10^{-7}) therefore predicts 13.2 chance cells across the survey, an expectation in cells and not in windows. The frozen products carry the observed count in the same units: 229 on-star cells reach 5σ, in 20 windows, and 174 of them belong to the four stage-1 flags, three of which are CO. The remaining 55 cells, over 16 windows, stand against the 13.2 expected, a factor 4.2 that is the product of two, only the second of them correlation. Correlation cannot move the expected *number* above 5σ, which is Np whatever the correlation; it moves the clumping. Summing $1 - (1 - 2.87 \times 10^{-7})^{n_{\rm cells}}$ over the 431 windows predicts 10.9 windows carrying at least one on-star crossing against the 16 observed, a factor 1.5 at Poisson $p = 0.09$; within a crossing window the iid expectation is 1.2 cells against 3.4 observed, a factor 2.8, and that one is channel adjacency. The two multiply to 4.2. The budget is also almost entirely fine-class: 4.58×10^7 of the 4.60×10^7 cells, so 10.9 chance crossings are expected among the 118 Class A windows against 0.03 among the 313 Class B ones, which is why all 20 crossings are fine windows.

The control ensembles calibrate this empirically, no Gaussian assumption surviving. The same arithmetic on each window's 512 controls predicts 134 of the 431 windows carrying at least one control above 5σ; 125 do, agreeing to 7 per cent. The observed per-window control maxima track the iid-Gaussian prediction for $512 \times n_{\rm cells}$ trials at a median ratio of 0.995 (5th–95th percentile 0.91–1.12), separately in both classes: fine 5.77 observed against 5.74 predicted, coarse 4.41 against 4.47. That agreement is partly bookkeeping: $n_{\rm cells}$ counts grid cells, not independent trials, and both axes are oversampled, in frequency by the Hanning correlation ($\rho_1 = 2/3$, Appendix A) and in drift by a grid whose adjacent trials differ by about half a channel over the track. Dividing the trial count by a combined factor of 4 lowers the predictions to 5.50 and 4.17, leaving a +5 and +6 per cent excess at the controls in the two classes independently. Mild non-Gaussianity of that size is exactly why the operative statement is the empirical rank and not the Gaussian budget. The residual window-level factor 1.5 at the star against 0.93 at the controls is the closest measurement of position-specific excess at the phase centre the release affords. It is marginal, and the crossing list is dominated by CO(2→1) at 230.5 GHz toward disc hosts, the expected astrophysical reading. The operative false-alarm statement remains the rank against the control ensemble. Beyond the shared primary-beam gain, neighbouring control positions are correlated below the synthesised-beam scale, so independent spatial trials fall from 512 to $N_{\rm eff} \simeq 30$–43, a floor to which the empirical 1.2 per cent rate, measured on the correlated ensembles themselves, is immune by construction. That figure is the compact case: taking the archive's own synthesised beam (0.10–5.7 arcsec, median 0.84) and $N_{\rm geom} = \pi(r_{\rm out}^2 - r_{\rm in}^2)/1.133\theta_{\rm beam}^2$ gives 36 to 1.4×10^5 resolution elements, median 995, across the 431 windows. That counts what the annulus *contains*, so it caps $N_{\rm eff}$ at the 512 positions actually drawn and shows the blanket 30–43 to be a worst case, not a typical one. The same metadata exposes two geometric limitations, unrepaired here because repairing them means re-running the extraction. 141 windows (33 per cent) come from 38 ACA 7 m execution blocks while $\theta_{\rm PB}$ is $1.22\,\lambda/12\,$m throughout, so there the annulus sits at a smaller fraction of the true primary beam than 0.14–0.78 implies and its gains are higher than quoted; and in 133 windows $r_{\rm in}$ falls below the synthesised beam, so the innermost controls are not resolution-independent of the star, which for a typical ACA geometry is ∼4 per cent of the 512, far below the median's breakdown point. Of the four stage-1 flags only HD 48370 Band 6 is ACA. Both consequences run in the conservative direction. On the true 7 m beam the ACA annuli span 0.08–0.46 of $\theta_{\rm PB}$ at a median area-weighted gain of 0.75 against 0.47 for the 12 m windows, so exchangeability in flux density improves there; and of the 41 windows carrying a primary-beam correction above 2 per cent, 36 are 12 m and 5 are 7 m, with all four at the 1.81× maximum 12 m, so the applied correction is overstated only for the five ACA rows, where the 1.71× dish-diameter ratio makes a ×1.05 correction really ×1.017.

No single stratum drives the rank-first rate: under the withdrawn asymmetric statistic it moved only between 11 and 20 per cent across band, channelisation and primary-beam offset, and disc- or planet-hosting stars showed a *lower* rate than non-hosts, so shared circumstellar emission does not drive it. Excluding the 15 β Pictoris windows leaves one unattributed crossing among 416, against 0.81 expected ($P(\geq 1) = 56$ per cent).

I. ANALYSIS AUDIT TRAIL: EXCLUDED ENTRIES AND THE RETIRED STATISTIC

Five pre-freeze changes are logged with their effect on released products in the repository change-log, and four are treated above. Only the replacement of the asymmetric region-max statistic by the symmetric single-position form of §4.1 (release re-run, criterion frozen) changes a released number. Counted *without* the $\geq 5\sigma$ amplitude gate, that is on rank alone, rank-first windows 65/431 (15.1 per cent) become 5/431 (1.2 per cent); counted *with* the gate, which is the stage-1 flag of the main text, 7 become 4 and the AU Mic crossing becomes unflagged. Under a pure-noise null the retired region maximum over $n_{\rm src}$ positions beats single controls with probability $n_{\rm src}/(n_{\rm src} + n_{\rm ctrl}) = 20.9$ per cent, the order of the measured region-max rate. Nothing derived from it enters an operative false-alarm probability.

The four classes of excluded entry. (i) Four target/bands (Barnard's Star [B7], Wolf 359 [B7], SCR J1845−6357, Wolf 358) have no carrier result from a process-level failure; each keeps a valid continuum non-detection and no EIRP threshold. (ii) Giclas 9−38A and Giclas 9−38B [B3] are excluded entirely, the pair's only matched MOUS pointing ∼52 arcmin from either star, 35× its own primary-beam FWHM; the carrier step aborts correctly, continuum imaging lacks that guard. (iii) Six windows across five stars (51 Eri, HD 10647, HD 139664, TWA 7, HD 92945 twice) fail the noise test of §5.2: all are 15.62-MHz coarse windows whose reported rms is inconsistent with their reported integration by three to four orders of magnitude and sits a factor 388–669 below the shallowest sibling of the same block and channel width, a normalisation signature, not weather or configuration (§5.2). (iv) The catalogue entry "γ Lupi" resolves to the F4V debris-disc host HD 139664 at 17.40 pc, not the naked-eye γ Lupi; all such entries are relabelled and no measured value changes. The supporting quality audit is clean across all 431 windows bar four χ^1 Ori Band 3 windows flagged as elevated (0.9–1.1× the Band 3 median, weakening their thresholds by $\lesssim$20 per cent in amplitude).

J. THE CONDITIONAL SEARCHED-DOMAIN TRANSMITTER FRACTION

No number here appears among the headline quantities of Table 4; the appendix is kept for the framework. A zero detection constrains a fraction of systems only through a per-system probability, factorised as

$$C_i(P) \;=\; F_i \; R_i(P) \; D_i \; C_{{\rm morph},i}, \tag{J1}$$

TABLE 11

Conditional searched-domain transmitter fraction, illustrative calculation only. *These numbers assume every searched star was broadcasting continuously, in band, at the stated power during the observation. They are not a demographic limit on any stellar population.* (a) the 95 per cent upper limit $f_{95}(P)$ above EIRP P, n the systems with non-zero completeness there; the *measured* column uses the injection-measured recovery, whose single-configuration transfer error is unquantified. (b) epoch-activity dependence at $P \geq 10^{16}$ W, thresholded and measured-completeness. 'Unconstrained' means no bound at that combination.

(a)

P (W)	thresholded (unit C)		measured C	
	n	f_{95}	n	f_{95}
3×10^{13}	4	52.7%	2	unconstrained
10^{14}	16	17.1%	11	47.1%
10^{15}	59	5.0%	47	9.5%
10^{16}	80	3.7%	58	6.3%
10^{17}	81	3.6%	59	6.0%

(b)

Epoch activity $p_{\rm epoch}$	f_{95} thresholded	f_{95} measured
1 (continuous)	3.7%	6.3%
0.5	6.5%	11.0%
0.1	29.2%	48.8%
0.01	unconstrained	unconstrained

with F_i the chance the transmitter's sky frequency lies in system i's searched intervals, $R_i(P)$ the chance a transmitter of EIRP P would have crossed that system's threshold, D_i the chance it emitted during the analysed archival interval, and $C_{{\rm morph},i}$ the measured recovery of a carrier of the searched drift and temporal morphology (Appendix B). The probability of zero credible detections if a fraction f of systems hosts one is

$$P(N_{\rm det} = 0 \,|\, f) = \prod_{i=1}^{N_{\rm sys}} (1 - f\, C_i)\,, \tag{J2}$$

and the one-sided 95 per cent upper limit f_{95} solves $P(N_{\rm det} = 0 \,|\, f_{95}) = 0.05$ over the 81 systems.

Everything below sets $F_i = D_i = 1$ *by conditioning, not by measurement*. At EIRP $\geq 10^{16}$ W the measured form gives $f_{95} \approx 6$ per cent over the 58 systems carrying a Class A window, with idealised brackets 3.7 and 8.8 per cent. Folding in the dwell-measured Class B curve, so each window carries the completeness measured for the morphology it searches, extends the calculation to 80 systems and moves the number to 4.3 per cent. Table 11 carries the ladder in EIRP and in epoch activity.

How far this travels is bounded three ways. The Class A recovery curve is measured on one configuration (AU Mic, 488-kHz Band 6) and applied to all 118 Class A windows; the $^{+16}_{-14}$ percentage-point binomial interval is sampling noise within that one run and does not bound the transfer error, whereas scaling the curve by 0.5–2× moves the number from 6.3 to 5.0–12.3 per cent, which is the honest width. The C_i are not independent: windows group into 102 execution blocks with an intraclass correlation of 0.24, whose design effect moves 6 to 7.1 per cent, an order below the transfer term. Under any transmitter-frequency prior spread across ALMA's tuning range the bound vanishes, the per-system Class A union being 1.73 GHz in the median against ∼75 GHz.

Writing $p_{\rm epoch}$ for the chance a transmitter is on in any one epoch, a system *searched* in n_i independent execution blocks is sampled n_i times, so $D_i = 1 - (1 - p_{\rm epoch})^{n_i}$; 18 of the 81 systems have $n_i > 1$, against 63 with two or more public blocks archived (§3), so the bound below is what the searched epochs support. With at most three epochs at year-scale separations this is a phase-averaged bound, and an intermittent transmitter is at least as plausible *a priori*, so the $p_{\rm epoch} = 1$ entries should never be quoted without it.

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